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## 14. Mean-Driven and Fluctuation-Driven Persistent Activity in Recurrent Networks

*Alfonso Renart, Rubén Moreno■Bote, Xiao■Jing Wang, Néstor Parga*

https://doi.org/10.1162/neco.2007.19.1.1 | pp. 1-46

Spike trains from cortical neurons show a high degree of irregularity, with coefficients of variation (CV) of their interspike interval (ISI) distribution close to or higher than one. It has been suggested that this irregularity might be a reflection of a particular dynamical state of the local cortical circuit in which excitation and inhibition balance each other. In this "balanced" state, the mean current to the neurons is below threshold, and firing is driven by current fluctuations, resulting in irregular Poisson-like spike trains. Recent data show that the degree of irregularity in neuronal spike trains recorded during the delay period of working memory experiments is the same for both low-activity states of a few Hz and for elevated, persistent activity states of a few tens of Hz. Since the difference between these persistent activity states cannot be due to external factors coming from sensory inputs, this suggests that the underlying network dynamics might support coexisting balanced states at different firing rates. We use mean field techniques to study the possible existence of multiple balanced steady states in recurrent networks of current-based leaky integrate-and-fire (LIF) neurons. To assess the degree of balance of a steady state, we extend existing mean-field theories so that not only the firing rate, but also the coefficient of variation of the interspike interval distribution of the neurons, are determined self-consistently. Depending on the connectivity parameters of the network, we find bistable solutions of different types. If the local recurrent connectivity is mainly excitatory, the two stable steady states differ mainly in the mean current to the neurons. In this case, the mean drive in the elevated persistent activity state is suprathreshold and typically characterized by low spiking irregularity. If the local recurrent excitatory and inhibitory drives are both large and nearly balanced, or even dominated by inhibition, two stable states coexist, both with subthreshold current drive. In this case, the spiking variability in both the resting state and the mnemonic persistent state is large, but the balance condition implies parameter fine-tuning. Since the degree of required fine-tuning increases with network size and, on the other hand, the size of the fluctuations in the afferent current to the cells increases for small networks, overall we find that fluctuation-driven persistent activity in the very simplified type of models we analyze is not a robust phenomenon. Possible implications of considering more realistic models are discussed.

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## 15. Exact Subthreshold Integration with Continuous Spike Times in Discrete-Time Neural Network Simulations

*Abigail Morrison, Sirko Straube, Hans Ekkehard Pleßer, Markus Diesmann*

https://doi.org/10.1162/neco.2007.19.1.47 | pp. 47-79

Very large networks of spiking neurons can be simulated efficiently in parallel under the constraint that spike times are bound to an equidistant time grid. Within this scheme, the subthreshold dynamics of a wide class of integrate-and-fire-type neuron models can be integrated exactly from one grid point to the next. However, the loss in accuracy caused by restricting spike times to the grid can have undesirable consequences, which has led to interest in interpolating spike times between the grid points to retrieve an adequate representation

of network dynamics. We demonstrate that the exact integration scheme can be combined naturally with off-grid spike events found by interpolation. We show that by exploiting the existence of a minimal synaptic propagation delay, the need for a central event queue is removed, so that the precision of event-driven simulation on the level of single neurons is combined with the efficiency of time-driven global scheduling. Further, for neuron models with linear subthreshold dynamics, even local event queuing can be avoided, resulting in much greater efficiency on the single-neuron level. These ideas are exemplified by two implementations of a widely used neuron model. We present a measure for the efficiency of network simulations in terms of their integration error and show that for a wide range of input spike rates, the novel techniques we present are both more accurate and faster than standard techniques.

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## 16. The Road to Chaos by Time-Asymmetric Hebbian Learning in Recurrent Neural Networks

*Colin Molter, Utku Salihoglu, Hugues Bersini*

https://doi.org/10.1162/neco.2007.19.1.80 | pp. 80-110

This letter aims at studying the impact of iterative Hebbian learning algorithms on the recurrent neural network's underlying dynamics. First, an iterative supervised learning algorithm is discussed. An essential improvement of this algorithm consists of indexing the attractor information items by means of external stimuli rather than by using only initial conditions, as Hopfield originally proposed. Modifying the stimuli mainly results in a change of the entire internal dynamics, leading to an enlargement of the set of attractors and potential memory bags. The impact of the learning on the network's dynamics is the following: the more information to be stored as limit cycle attractors of the neural network, the more chaos prevails as the background dynamical regime of the network. In fact, the background chaos spreads widely and adopts a very unstructured shape similar to white noise. Next, we introduce a new form of supervised learning that is more plausible from a biological point of view: the network has to learn to react to an external stimulus by cycling through a sequence that is no longer specified a priori. Based on its spontaneous dynamics, the network decides "on its own" the dynamical patterns to be associated with the stimuli. Compared with classical supervised learning, huge enhancements in storing capacity and computational cost have been observed. Moreover, this new form of supervised learning, by being more "respectful" of the network intrinsic dynamics, maintains much more structure in the obtained chaos. It is still possible to observe the traces of the learned attractors in the chaotic regime. This complex but still very informative regime is referred to as "frustrated chaos."

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## 17. Analysis and Design of Echo State Networks

*Mustafa C. Ozturk, Dongming Xu, José C. Pr■■ncipe*

https://doi.org/10.1162/neco.2007.19.1.111 | pp. 111-138

The design of echo state network (ESN) parameters relies on the selection of the maximum eigenvalue of the linearized system around zero (spectral radius). However, this procedure does not quantify in a systematic manner the performance of the ESN in terms of approximation error. This article presents a functional space approximation framework to better understand the operation of ESNs and proposes an information-theoretic metric, the average entropy of echo states, to assess the richness of the ESN dynamics. Furthermore, it provides an interpretation of the ESN dynamics rooted in system theory as

families of coupled linearized systems whose poles move according to the input signal dynamics. With this interpretation, a design methodology for functional approximation is put forward where ESNs are designed with uniform pole distributions covering the frequency spectrum to abide by the richness metric, irrespective of the spectral radius. A single bias parameter at the ESN input, adapted with the modeling error, configures the ESN spectral radius to the input-output joint space. Function approximation examples compare the proposed design methodology versus the conventional design.

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## 18. Invariant Global Motion Recognition in the Dorsal Visual System: A Unifying Theory

*Edmund T. Rolls, Simon M. Stringer*

https://doi.org/10.1162/neco.2007.19.1.139 | pp. 139-169

The motion of an object (such as a wheel rotating) is seen as consistent independent of its position and size on the retina. Neurons in higher cortical visual areas respond to these global motion stimuli invariantly, but neurons in early cortical areas with small receptive fields cannot represent this motion, not only because of the aperture problem but also because they do not have invariant representations. In a unifying hypothesis with the design of the ventral cortical visual system, we propose that the dorsal visual system uses a hierarchical feedforward network architecture (V1, V2, MT, MSTd, parietal cortex) with training of the connections with a short-term memory trace associative synaptic modification rule to capture what is invariant at each stage. Simulations show that the proposal is computationally feasible, in that invariant representations of the motion flow fields produced by objects self-organize in the later layers of the architecture. The model produces invariant representations of the motion flow fields produced by global in-plane motion of an object, in-plane rotational motion, looming versus receding of the object, and object-based rotation about a principal axis. Thus, the dorsal and ventral visual systems may share some similar computational principles.

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## 19. Recurrent Cerebellar Loops Simplify Adaptive Control of Redundant and Nonlinear Motor Systems

*John Porrill, Paul Dean*

https://doi.org/10.1162/neco.2007.19.1.170 | pp. 170-193

We have described elsewhere an adaptive filter model of cerebellar learning in which the cerebellar microcircuit acts to decorrelate motor commands from their sensory consequences (Dean, Porrill, & Stone, 2002). Learning stability required the cerebellar microcircuit to be embedded in a recurrent loop, and this has been shown to lead to a simple and modular adaptive control architecture when applied to the linearized 3D vestibular ocular reflex (Porrill, Dean, & Stone, 2004). Here we investigate the properties of recurrent loop connectivity in the case of redundant and nonlinear motor systems and illustrate them using the example of kinematic control of a simulated two-joint robot arm. We demonstrate that (1) the learning rule does not require unavailable motor error signals or complex neural reference structures to estimate such signals (i.e., it solves the motor error problem) and (2) control of redundant systems is not subject to the nonconvexity problem in which incorrect average motor commands are learned for end-effector positions that can be accessed in more than one arm configuration. These properties suggest a central functional role for the closed cerebellar loops, which have been shown to be ubiquitous in motor systems (e.g., Kelly &

Strick, 2003).

## 20. Free-Lunch Learning: Modeling Spontaneous Recovery of Memory

*James V. Stone, Peter E. Jupp*

https://doi.org/10.1162/neco.2007.19.1.194 | pp. 194-217

After a language has been learned and then forgotten, relearning some words appears to facilitate spontaneous recovery of other words. More generally, relearning partially forgotten associations induces recovery of other associations in humans, an effect we call free-lunch learning (FLL). Using neural network models, we prove that FLL is a necessary consequence of storing associations as distributed representations. Specifically, we prove that (1) FLL becomes increasingly likely as the number of synapses (connection weights) increases, suggesting that FLL contributes to memory in neurophysiological systems, and (2) the magnitude of FLL is greatest if inactive synapses are removed, suggesting a computational role for synaptic pruning in physiological systems. We also demonstrate that FLL is different from generalization effects conventionally associated with neural network models. As FLL is a generic property of distributed representations, it may constitute an important factor in human memory.

## 21. Linear Multilayer ICA Generating Hierarchical Edge Detectors

*Yoshitatsu Matsuda, Kazunori Yamaguchi*

https://doi.org/10.1162/neco.2007.19.1.218 | pp. 218-230

In this letter, a new ICA algorithm, linear multilayer ICA (LMICA), is proposed. There are two phases in each layer of LMICA. One is the mapping phase, where a two-dimensional mapping is formed by moving more highly correlated (nonindependent) signals closer with the stochastic multidimensional scaling network. Another is the local-ICA phase, where each neighbor (namely, highly correlated) pair of signals in the mapping is separated by MaxKurt algorithm. Because in LMICA only a small number of highly correlated pairs have to be separated, it can extract edge detectors efficiently from natural scenes. We conducted numerical experiments and verified that LMICA generates hierarchical edge detectors from large-size natural scenes.

## 22. Functional Network Topology Learning and Sensitivity Analysis Based on ANOVA Decomposition

*Enrique Castillo, Noelia Sánchez■Maroño, Amparo Alonso■Betanzos, Carmen Castillo*

https://doi.org/10.1162/neco.2007.19.1.231 | pp. 231-257

A new methodology for learning the topology of a functional network from data, based on the ANOVA decomposition technique, is presented. The method determines sensitivity (importance) indices that allow a decision to be made as to which set of interactions among variables is relevant and which is irrelevant to the problem under study. This immediately suggests the network topology to be used in a given problem. Moreover, local sensitivities to small changes in the data can be easily calculated. In this way, the dual optimization problem gives the local sensitivities. The methods are illustrated by their application to artificial and real examples.

## 23. Second-Order Cone Programming Formulations for Robust Multiclass Classification

*Ping Zhong, Masao Fukushima*

https://doi.org/10.1162/neco.2007.19.1.258 | pp. 258-282

Multiclass classification is an important and ongoing research subject in machine learning. Current support vector methods for multiclass classification implicitly assume that the parameters in the optimization problems are known exactly. However, in practice, the parameters have perturbations since they are estimated from the training data, which are usually subject to measurement noise. In this article, we propose linear and nonlinear robust formulations for multiclass classification based on the M-SVM method. The preliminary numerical experiments confirm the robustness of the proposed method.

## 24. Fast Generalized Cross-Validation Algorithm for Sparse Model Learning

*S. Sundararajan, Shirish Shevade, S. Sathiya Keerthi*

https://doi.org/10.1162/neco.2007.19.1.283 | pp. 283-301

We propose a fast, incremental algorithm for designing linear regression models. The proposed algorithm generates a sparse model by optimizing multiple smoothing parameters using the generalized cross-validation approach. The performances on synthetic and real-world data sets are compared with other incremental algorithms such as Tipping and Faul's fast relevance vector machine, Chen et al.'s orthogonal least squares, and Orr's regularized forward selection. The results demonstrate that the proposed algorithm is competitive.

## 25. Addendum

*(authors unknown)*

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No abstract available.

## 26. Receptive Field Structures for Recognition

*Benjamin Balas, Pawan Sinha*

https://doi.org/10.1162/089976606775623315 | pp. 497-520

Localized operators, like Gabor wavelets and difference-of-gaussian filters, are considered useful tools for image representation. This is due to their ability to form a sparse code that can serve as a basis set for high-fidelity reconstruction of natural images. However, for many visual tasks, the more appropriate criterion of representational efficacy is recognition rather than reconstruction. It is unclear whether simple local features provide the stability necessary to subserve robust recognition of complex objects. In this article, we search the space of two-lobed differential operators for those that constitute a good representational code under recognition and discrimination criteria. We find that a novel operator, which we call the dissociated dipole, displays useful properties in this regard. We describe simple computational experiments to assess

the merits of such dipoles relative to the more traditional local operators. The results suggest that nonlocal operators constitute a vocabulary that is stable across a range of image transformations.

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## 27. Receptive Field Structures for Recognition

*Benjamin Balas, Pawan Sinha*

https://doi.org/10.1162/neco.2006.18.3.497 | pp. 497-520

Localized operators, like Gabor wavelets and difference-of-gaussian filters, are considered useful tools for image representation. This is due to their ability to form a sparse code that can serve as a basis set for high-fidelity reconstruction of natural images. However, for many visual tasks, the more appropriate criterion of representational efficacy is recognition rather than reconstruction. It is unclear whether simple local features provide the stability necessary to subserve robust recognition of complex objects. In this article, we search the space of two-lobed differential operators for those that constitute a good representational code under recognition and discrimination criteria. We find that a novel operator, which we call the dissociated dipole, displays useful properties in this regard. We describe simple computational experiments to assess the merits of such dipoles relative to the more traditional local operators. The results suggest that nonlocal operators constitute a vocabulary that is stable across a range of image transformations.

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## 28. A Neural Model of the Scintillating Grid Illusion: Disinhibition and Self-Inhibition in Early Vision

*Yingwei Yu, Yoonsuck Choe*

https://doi.org/10.1162/089976606775623306 | pp. 521-544

A stationary display of white discs positioned on intersecting gray bars on a dark background gives rise to a striking scintillating effect--the scintillating grid illusion. The spatial and temporal properties of the illusion are well known, but a neuronal-level explanation of the mechanism has not been fully investigated. Motivated by the neurophysiology of the Limulus retina, we propose disinhibition and self-inhibition as possible neural mechanisms that may give rise to the illusion. In this letter, a spatiotemporal model of the early visual pathway is derived that explicitly accounts for these two mechanisms. The model successfully predicted the change of strength in the illusion under various stimulus conditions, indicating that low-level mechanisms may well explain the scintillating effect in the illusion.

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A stationary display of white discs positioned on intersecting gray bars on a dark background gives rise to a striking scintillating effect—the scintillating grid illusion. The spatial and temporal properties of the illusion are well known, but a neuronal-level explanation of the mechanism has not been fully investigated. Motivated by the neurophysiology of the Limulus retina, we propose disinhibition and self-inhibition as possible neural mechanisms that may give rise to the illusion. In this letter, a spatiotemporal model of the

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## 30. A Comparison of Descriptive Models of a Single Spike Train by Information-Geometric Measure

*Hiroyuki Nakahara, ■■■-■■■ ■■■■■, Barry J. Richmond*

https://doi.org/10.1162/089976606775623289 | pp. 545-568

In examining spike trains, different models are used to describe their structure. The different models often seem quite similar, but because they are cast in different formalisms, it is often difficult to compare their predictions. Here we use the information-geometric measure, an orthogonal coordinate representation of point processes, to express different models of stochastic point processes in a common coordinate system. Within such a framework, it becomes straightforward to visualize higher-order correlations of different models and thereby assess the differences between models. We apply the information-geometric measure to compare two similar but not identical models of neuronal spike trains: the inhomogeneous Markov and the mixture of Poisson models. It is shown that they differ in the second- and higher-order interaction terms. In the mixture of Poisson model, the second- and higher-order interactions are of comparable magnitude within each order, whereas in the inhomogeneous Markov model, they have alternating signs over different orders. This provides guidance about what measurements would effectively separate the two models. As newer models are proposed, they also can be compared to these models using information geometry.

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## 32. A Theoretical Analysis of the Influence of Fixational Instability on the Development of Thalamocortical Connectivity

*Antonino Casile, Michele Rucci*

https://doi.org/10.1162/neco.2006.18.3.569 | pp. 569-590

Under natural viewing conditions, the physiological instability of visual fixation keeps the projection of the stimulus on the retina in constant motion. After eye opening, chronic exposure to a constantly moving retinal image might influence the experience-dependent refinement of cell response characteristics. The results of previous modeling studies have suggested a contribution of fixational instability to the Hebbian maturation of the receptive fields of V1 simple cells (Rucci, Edelman, & Wray, 2000; Rucci & Casile, 2004). This letter examines the origins of such a contribution. Using quasilinear models of lateral geniculate nucleus units and V1 simple cells, we derive analytical expressions for the second- order statistics of thalamocortical activity before and after eye opening. We show that in the presence of natural stimulation, fixational instability introduces a spatially uncorrelated signal in the retinal input, which strongly influences the structure of correlated activity in the model. This input signal produces a regime of thalamocortical activity similar to that present before eye opening and compatible with the Hebbian maturation of cortical receptive fields.

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## 33. A Theoretical Analysis of the Influence of Fixational Instability on the Development of Thalamocortical Connectivity

*Antonino Casile, Michele Rucci*

https://doi.org/10.1162/089976606775623270 | pp. 569-590

Under natural viewing conditions, the physiological instability of visual fixation keeps the projection of the stimulus on the retina in constant motion. After eye opening, chronic exposure to a constantly moving retinal image might influence the experience-dependent refinement of cell response characteristics. The results of previous modeling studies have suggested a contribution of fixational instability to the Hebbian maturation of the receptive fields of V1 simple cells (Rucci, Edelman, & Wray, 2000; Rucci & Casile, 2004). This letter examines the origins of such a contribution. Using quasilinear models of lateral geniculate nucleus units and V1 simple cells, we derive analytical expressions for the second-order statistics of thalamocortical activity before and after eye opening. We show that in the presence of natural stimulation, fixational instability introduces a spatially uncorrelated signal in the retinal input, which strongly influences the structure of correlated activity in the model. This input signal produces a regime of thalamocortical activity similar to that present before eye opening and compatible with the Hebbian maturation of cortical receptive fields.

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## 34. Learning Beyond Finite Memory in Recurrent Networks of Spiking Neurons

*Peter Ti■o, Ashely J. S. Mills*

https://doi.org/10.1162/089976606775623360 | pp. 591-613

We investigate possibilities of inducing temporal structures without fading memory in recurrent networks of spiking neurons strictly operating in the pulse-coding regime. We extend the existing gradient-based algorithm for training feedforward spiking neuron networks, SpikeProp (Bohte, Kok, & La Poutré, 2002), to

recurrent network topologies, so that temporal dependencies in the input stream are taken into account. It is shown that temporal structures with unbounded input memory specified by simple Moore machines (MM) can be induced by recurrent spiking neuron networks (RSNN). The networks are able to discover pulse-coded representations of abstract information processing states coding potentially unbounded histories of processed inputs. We show that it is often possible to extract from trained RSNN the target MM by grouping together similar spike trains appearing in the recurrent layer. Even when the target MM was not perfectly induced in a RSNN, the extraction procedure was able to reveal weaknesses of the induced mechanism and the extent to which the target machine had been learned.

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## 36. Effects of Fast Presynaptic Noise in Attractor Neural Networks

*Jesús M. Cortés, Joaqu■■n J. Torres, Joaquín Marro, Pedro L. Garrido, Hilbert J. Kappen*

https://doi.org/10.1162/089976606775623342 | pp. 614-633

We study both analytically and numerically the effect of presynaptic noise on the transmission of information in attractor neural networks. The noise occurs on a very short timescale compared to that for the neuron dynamics and it produces short-time synaptic depression. This is inspired in recent neurobiological findings that show that synaptic strength may either increase or decrease on a short timescale depending on presynaptic activity. We thus describe a mechanism by which fast presynaptic noise enhances the neural network sensitivity to an external stimulus. The reason is that, in general, presynaptic noise induces nonequilibrium behavior and, consequently, the space of fixed points is qualitatively modified in such a way that the system can easily escape from the attractor. As a result, the model shows, in addition to pattern recognition, class identification and categorization, which may be relevant to the understanding of some of the brain complex tasks.

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## 38. Response Variability in Balanced Cortical Networks

*Alexander Lerchner, Cristina Ursta, John Hertz, Mandana Ahmadi, Pauline Ruffiot, Søren Enemark*

https://doi.org/10.1162/neco.2006.18.3.634 | pp. 634-659

We study the spike statistics of neurons in a network with dynamically balanced excitation and inhibition. Our model, intended to represent a generic cortical column, comprises randomly connected excitatory and inhibitory leaky integrate-and-fire neurons, driven by excitatory input from an external population. The high connectivity permits a mean field description in which synaptic currents can be treated as gaussian noise, the mean and autocorrelation function of which are calculated self-consistently from the firing statistics of single model neurons. Within this description, a wide range of Fano factors is possible. We find that the irregularity of spike trains is controlled mainly by the strength of the synapses relative to the difference between the firing threshold and the postfiring reset level of the membrane potential. For moderately strong synapses, we find spike statistics very similar to those observed in primary visual cortex.

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*Alexander Lerchner, Cristina Ursta, John Hertz, Mandana Ahmadi, Pauline Ruffiot, Søren Enemark*

https://doi.org/10.1162/089976606775623261 | pp. 634-659

We study the spike statistics of neurons in a network with dynamically balanced excitation and inhibition. Our model, intended to represent a generic cortical column, comprises randomly connected excitatory and inhibitory leaky integrate-and-fire neurons, driven by excitatory input from an external population. The high connectivity permits a mean field description in which synaptic currents can be treated as gaussian noise, the mean and autocorrelation function of which are calculated self-consistently from the firing statistics of single model neurons. Within this description, a wide range of Fano factors is possible. We find that the irregularity of spike trains is controlled mainly by the strength of the synapses relative to the difference between the firing threshold and the postfiring reset level of the membrane potential. For moderately strong synapses, we find spike statistics very similar to those observed in primary visual cortex.

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## 40. The Costs of Ignoring High-Order Correlations in Populations of Model Neurons

*Melchi M. Michel, Robert A. Jacobs*

https://doi.org/10.1162/089976606775623298 | pp. 660-682

Investigators debate the extent to which neural populations use pair-wise and higher-order statistical dependencies among neural responses to represent information about a visual stimulus. To study this issue, three statistical decoders were used to extract the information in the responses of model neurons about the binocular disparities present in simulated pairs of left-eye and right-eye images: (1) the full joint probability decoder considered all possible statistical relations among neural responses as potentially important; (2) the dependence tree decoder also considered all possible relations as potentially important, but it approximated high-order statistical correlations using a computationally tractable procedure; and (3) the independent response decoder, which assumed that neural responses are statistically independent, meaning that all correlations should be zero and thus can be ignored. Simulation results indicate that high-order correlations among model neuron responses contain significant information about binocular disparities and that the amount of this high-order information increases rapidly as a function of neural population size. Furthermore, the results highlight the potential importance of the dependence tree decoder to neuroscientists as a powerful but still practical way of approximating high-order correlations among neural responses.

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Investigators debate the extent to which neural populations use pairwise and higher-order statistical dependencies among neural responses to represent information about a visual stimulus. To study this issue, three statistical decoders were used to extract the information in the responses of model neurons about the binocular disparities present in simulated pairs of left-eye and right-eye images: (1) the full joint probability decoder considered all possible statistical relations among neural responses as potentially important; (2) the dependence tree decoder also considered all possible relations as potentially important, but it approximated high-order statistical correlations using a computationally tractable procedure; and (3) the independent response decoder, which assumed that neural responses are statistically independent, meaning that all correlations should be zero and thus can be ignored. Simulation results indicate that high-order correlations among model neuron responses contain significant information about binocular disparities and that the amount of this high-order information increases rapidly as a function of neural population size. Furthermore, the results highlight the potential importance of the dependence tree decoder to neuroscientists as a powerful but still practical way of approximating high-order correlations among neural responses.

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## 42. Dynamical Behaviors of Delayed Neural Network Systems with Discontinuous Activation Functions

*Wenlian Lu, Tianping Chen*

https://doi.org/10.1162/neco.2006.18.3.683 | pp. 683-708

In this letter, without assuming the boundedness of the activation functions, we discuss the dynamics of a class of delayed neural networks with discontinuous activation functions. A relaxed set of sufficient conditions is derived, guaranteeing the existence, uniqueness, and global stability of the equilibrium point. Convergence behaviors for both state and output are discussed. The constraints imposed on the feedback matrix are independent of the delay parameter and can be validated by the linear matrix inequality technique. We also prove that the solution of delayed neural networks with discontinuous activation functions can be regarded as a limit of the solutions of delayed neural networks with high-slope continuous activation functions.

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## 43. Dynamical Behaviors of Delayed Neural Network Systems with Discontinuous Activation Functions

*Wenlian Lu, Tianping Chen*

https://doi.org/10.1162/089976606775623324 | pp. 683-708

In this letter, without assuming the boundedness of the activation functions, we discuss the dynamics of a class of delayed neural networks with discontinuous activation functions. A relaxed set of sufficient conditions is derived, guaranteeing the existence, uniqueness, and global stability of the equilibrium point. Convergence behaviors for both state and output are discussed. The constraints imposed on the feedback matrix are independent of the delay parameter and can be validated by the linear matrix inequality technique. We also prove that the solution of delayed neural networks with discontinuous activation functions can be regarded as a limit of the solutions of delayed neural networks with high-slope continuous activation functions.

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## 44. A Generalized Contrast Function and Stability Analysis for Overdetermined Blind Separation of Instantaneous Mixtures

*Xiaolong Zhu, Xian■Da Zhang, Jimin Ye*

https://doi.org/10.1162/089976606775623333 | pp. 709-728

In this letter, the problem of blind separation of n independent sources from their m linear instantaneous mixtures is considered. First, a generalized contrast function is defined as a valuable extension of the existing classical and nonsymmetrical contrast functions. It is applicable to the overdetermined blind separation (m > n) with an unknown number of sources, because not only independent components but also redundant ones are allowed in the outputs of a separation system. Second, a natural gradient learning algorithm developed primarily for the complete case (m = n) is shown to work as well with an n × m or m × m separating matrix, for each optimizes a certain mutual information contrast function. Finally, we present stability analysis for a newly proposed generalized orthogonal natural gradient algorithm (which can perform the overdetermined blind separation when n is unknown), obtaining an expectable result that its local stability conditions are slightly stricter than those of the conventional natural gradient algorithm using an invertible mixing matrix (m = n).

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## 46. Connection and Coordination: The Interplay Between Architecture and Dynamics in Evolved Model Pattern Generators

*Sean Psujek, Jeffrey C. Ames, Randall D. Beer*

https://doi.org/10.1162/089976606775623351 | pp. 729-747

We undertake a systematic study of the role of neural architecture in shaping the dynamics of evolved model pattern generators for a walking task. First, we consider the minimum number of connections necessary to achieve high performance on this task. Next, we identify architectural motifs associated with high fitness. We then examine how high-fitness architectures differ in their ability to evolve. Finally, we demonstrate the existence of distinct parameter subgroups in some architectures and show that these subgroups are characterized by differences in neuron excitabilities and connection signs.

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## 47. Connection and Coordination: The Interplay Between Architecture and Dynamics in Evolved Model Pattern Generators

*Sean Psujek, Jeffrey C. Ames, Randall D. Beer*

https://doi.org/10.1162/neco.2006.18.3.729 | pp. 729-747

We undertake a systematic study of the role of neural architecture in shaping the dynamics of evolved model pattern generators for a walking task. First, we consider the minimum number of connections necessary to achieve high performance on this task. Next, we identify architectural motifs associated with high fitness. We then examine how high-fitness architectures differ in their ability to evolve. Finally, we demonstrate the existence of distinct parameter subgroups in some architectures and show that these subgroups are characterized by differences in neuron excitabilities and connection signs.

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## 48. An Invariance Property of Predictors in Kernel-Induced Hypothesis Spaces

*Nicola Ancona, Sebastiano Stramaglia*

https://doi.org/10.1162/neco.2006.18.4.749 | pp. 749-759

We consider kernel-based learning methods for regression and analyze what happens to the risk minimizer when new variables, statistically independent of input and target variables, are added to the set of input variables. This problem arises, for example, in the detection of causality relations between two time series. We find that the risk minimizer remains unchanged if we constrain the risk minimization to hypothesis spaces induced by suitable kernel functions. We show that not all kernel-induced hypothesis spaces enjoy this property. We present sufficient conditions ensuring that the risk minimizer does not change and show that they hold for inhomogeneous polynomial and gaussian radial basis function kernels. We also provide examples of kernel-induced hypothesis spaces whose risk minimizer changes if independent variables are added as input.

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## 49. An Invariance Property of Predictors in Kernel-Induced Hypothesis Spaces

*Nicola Ancona, Sebastiano Stramaglia*

https://doi.org/10.1162/089976606775774660 | pp. 749-759

We consider kernel-based learning methods for regression and analyze what happens to the risk minimizer when new variables, statistically independent of input and target variables, are added to the set of input variables. This problem arises, for example, in the detection of causality relations between two time series. We find that the risk minimizer remains unchanged if we constrain the risk minimization to hypothesis spaces induced by suitable kernel functions. We show that not all kernel-induced hypothesis spaces enjoy this property. We present sufficient conditions ensuring that the risk minimizer does not change and show that they hold for inhomogeneous polynomial and gaussian radial basis function kernels. We also provide examples of kernel-induced hypothesis spaces whose risk minimizer changes if independent variables are added as input.

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## 50. Modeling Sensorimotor Learning with Linear Dynamical Systems

*Sen Cheng, Philip N. Sabes*

https://doi.org/10.1162/neco.2006.18.4.760 | pp. 760-793

Recent studies have employed simple linear dynamical systems to model trial-by-trial dynamics in various sensorimotor learning tasks. Here we explore the theoretical and practical considerations that arise when employing the general class of linear dynamical systems (LDS) as a model for sensorimotor learning. In this framework, the state of the system is a set of parameters that define the current sensorimotor transformation— the function that maps sensory inputs to motor outputs. The class of LDS models provides a first-order approximation for any Markovian (state-dependent) learning rule that specifies the changes in the sensorimotor transformation that result from sensory feedback on each movement. We show that modeling the trial-by-trial dynamics of learning provides a sub-stantially enhanced picture of the process of adaptation compared to measurements of the steady state of adaptation derived from more traditional

blocked-exposure experiments. Specifically, these models can be used to quantify sensory and performance biases, the extent to which learned changes in the sensorimotor transformation decay over time, and the portion of motor variability due to either learning or performance variability. We show that previous attempts to fit such models with linear regression have not generally yielded consistent parameter estimates. Instead, we present an expectation-maximization algorithm for fitting LDS models to experimental data and describe the difficulties inherent in estimating the parameters associated with feedback-driven learning. Finally, we demonstrate the application of these methods in a simple sensorimotor learning experiment: adaptation to shifted visual feedback during reaching.

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## 51. Modeling Sensorimotor Learning with Linear Dynamical Systems

*Sen Cheng, Philip N. Sabes*

https://doi.org/10.1162/089976606775774651 | pp. 760-793

Recent studies have employed simple linear dynamical systems to model trial-by-trial dynamics in various sensorimotor learning tasks. Here we explore the theoretical and practical considerations that arise when employing the general class of linear dynamical systems (LDS) as a model for sensorimotor learning. In this framework, the state of the system is a set of parameters that define the current sensorimotor transformation-the function that maps sensory inputs to motor outputs. The class of LDS models provides a first-order approximation for any Markovian (state-dependent) learning rule that specifies the changes in the sensorimotor transformation that result from sensory feedback on each movement. We show that modeling the trial-by-trial dynamics of learning provides a substantially enhanced picture of the process of adaptation compared to measurements of the steady state of adaptation derived from more traditional blocked-exposure experiments. Specifically, these models can be used to quantify sensory and performance biases, the extent to which learned changes in the sensorimotor transformation decay over time, and the portion of motor variability due to either learning or performance variability. We show that previous attempts to fit such models with linear regression have not generally yielded consistent parameter estimates. Instead, we present an expectation-maximization algorithm for fitting LDS models to experimental data and describe the difficulties inherent in estimating the parameters associated with feedback-driven learning. Finally, we demonstrate the application of these methods in a simple sensorimotor learning experiment: adaptation to shifted visual feedback during reaching.

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## 52. Changing Roles for Temporal Representation of Odorant During the Oscillatory Response of the Olfactory Bulb

*Soyoun Kim, Benjamin H. Singer, Micha■ ■ochowski*

https://doi.org/10.1162/089976606775774697 | pp. 794-816

It has been hypothesized that the brain uses combinatorial as well as temporal coding strategies to represent stimulus properties. The mechanisms and properties of the temporal coding remain undetermined, although it has been postulated that oscillations can mediate formation of this type of code. Here we use a generic model of the vertebrate olfactory bulb to explore the possible role of oscillatory behavior in temporal coding. We show that three mechanisms--synaptic inhibition, slow self-inhibition and input properties--mediate formation of a temporal sequence of simultaneous activations of glomerular modules associated with specific odorants within the oscillatory response. The sequence formed depends on the

relative properties of odorant features and thus may mediate discrimination of odorants activating overlapping sets of glomeruli. We suggest that period-doubling transitions may be driven through excitatory feedback from a portion of the olfactory network acting as a coincidence modulator. Furthermore, we hypothesize that the period-doubling transition transforms the temporal code from a roster of odorant components to a signal of odorant identity and facilitates discrimination of individual odorants within mixtures.

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## 53. Changing Roles for Temporal Representation of Odorant During the Oscillatory Response of the Olfactory Bulb

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## 54. Computation of the Phase Response Curve: A Direct Numerical Approach

*W. Govaerts, Bart Sautois*

https://doi.org/10.1162/089976606775774688 | pp. 817-847

Neurons are often modeled by dynamical systems--parameterized systems of differential equations. A typical behavioral pattern of neurons is periodic spiking; this corresponds to the presence of stable limit cycles in the dynamical systems model. The phase resetting and phase response curves (PRCs) describe the reaction of the spiking neuron to an input pulse at each point of the cycle. We develop a new method for computing these curves as a by-product of the solution of the boundary value problem for the stable limit cycle. The method is mathematically equivalent to the adjoint method, but our implementation is computationally much faster and more robust than any existing method. In fact, it can compute PRCs even where the limit cycle can hardly be found by time integration, for example, because it is close to another stable limit cycle. In addition, we obtain the discretized phase response curve in a form that is ideally suited for most applications. We present several examples and provide the implementation in a freely available Matlab code.

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## 56. Multiperiodicity and Exponential Attractivity Evoked by Periodic External Inputs in Delayed Cellular Neural Networks

*Zhigang Zeng, Jun Wang*

https://doi.org/10.1162/neco.2006.18.4.848 | pp. 848-870

We show that an n-neuron cellular neural network with time-varying delay can have 2 n periodic orbits located in saturation regions and these periodic orbits are locally exponentially attractive. In addition, we give some conditions for ascertaining periodic orbits to be locally or globally exponentially attractive and allow them to locate in any designated region. As a special case of exponential periodicity, exponential stability of delayed cellular neural networks is also characterized. These conditions improve and extend the existing results in the literature. To illustrate and compare the results, simulation results are discussed in three numerical examples.

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## 57. Multiperiodicity and Exponential Attractivity Evoked by Periodic External Inputs in Delayed Cellular Neural Networks

*Zhigang Zeng, Jun Wang*

https://doi.org/10.1162/089976606775774624 | pp. 848-870

We show that an n-neuron cellular neural network with time-varying delay can have 2(n) periodic orbits located in saturation regions and these periodic orbits are locally exponentially attractive. In addition, we give some conditions for ascertaining periodic orbits to be locally or globally exponentially attractive and allow them to locate in any designated region. As a special case of exponential periodicity, exponential stability of delayed cellular neural networks is also characterized. These conditions improve and extend the existing results in the literature. To illustrate and compare the results, simulation results are discussed in three numerical examples.

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## 58. Smooth Gradient Representations as a Unifying Account of Chevreul's Illusion, Mach Bands, and a Variant of the Ehrenstein Disk

*Matthias S. Keil*

https://doi.org/10.1162/089976606775774705 | pp. 871-903

Recent evidence suggests that the primate visual system generates representations for object surfaces (where we consider representations for the surface attribute brightness). Object recognition can be expected to perform robustly if those representations are invariant despite environmental changes (e.g., in illumination). In real-world scenes, it happens, however, that surfaces are often overlaid by luminance gradients, which we define as smooth variations in intensity. Luminance gradients encode highly variable information, which may represent surface properties (curvature), nonsurface properties (e.g., specular highlights, cast shadows, illumination inhomogeneities), or information about depth relationships (cast shadows, blur). We argue, on grounds of the unpredictable nature of luminance gradients, that the visual system should establish corresponding representations, in addition to surface representations. We accordingly present a neuronal architecture, the so-called gradient system, which clarifies how spatially accurate gradient representations can be obtained by relying on only high-resolution retinal responses. Although the gradient system was designed and optimized for segregating, and generating, representations of luminance gradients with real-world luminance images, it is capable of quantitatively predicting psychophysical data on both Mach bands and Chevreul's illusion. It furthermore accounts qualitatively for a modified Ehrenstein disk.

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## 59. Smooth Gradient Representations as a Unifying Account of Chevreul's Illusion, Mach Bands, and a Variant of the Ehrenstein Disk

*Matthias S. Keil*

https://doi.org/10.1162/neco.2006.18.4.871 | pp. 871-903

Recent evidence suggests that the primate visual system generates representations for object surfaces (where we consider representations for the surface attribute brightness). Object recognition can be expected to perform robustly if those representations are invariant despite environmental changes (e.g., in illumination). In real-world scenes, it happens, however, that surfaces are often overlaid by luminance gradients, which we define as smooth variations in intensity. Luminance gradients encode highly variable information, which may represent surface properties (curvature), nonsurface properties (e.g., specular highlights, cast shadows, illumination inhomogeneities), or information about depth relationships (cast shadows, blur). We argue, on grounds of the unpredictable nature of luminance gradients, that the visual system should establish corresponding representations, in addition to surface representations. We accordingly present a neuronal architecture, the so-called gradient system, which clarifies how spatially accurate gradient representations can be obtained by relying on only high-resolution retinal responses. Although the gradient system was designed and optimized for segregating, and generating, representations of luminance gradients with real-world luminance images, it is capable of quantitatively predicting psychophysical data on both Mach bands and Chevreul's illusion. It furthermore accounts qualitatively for a modified Ehrenstein disk.

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## 60. Memory Capacity for Sequences in a Recurrent Network with Biological Constraints

*Christian Leibold, Richard Kempter*

https://doi.org/10.1162/neco.2006.18.4.904 | pp. 904-941

The CA3 region of the hippocampus is a recurrent neural network that is essential for the storage and replay of sequences of patterns that represent behavioral events. Here we present a theoretical framework to calculate a sparsely connected network's capacity to store such sequences. As in CA3, only a limited subset of neurons in the network is active at any one time, pattern retrieval is subject to error, and the resources for plasticity are limited. Our analysis combines an analytical mean field approach, stochastic dynamics, and cellular simulations of a time-discrete McCulloch-Pitts network with binary synapses. To maximize the number of sequences that can be stored in the network, we concurrently optimize the number of active neurons, that is, pattern size, and the firing threshold. We find that for one-step associations (i.e., minimal sequences), the optimal pattern size is inversely proportional to the mean connectivity c, whereas the optimal firing threshold is independent of the connectivity. If the number of synapses per neuron is fixed, the maximum number P of stored sequences in a sufficiently large, nonmodular network is independent of its number N of cells. On the other hand, if the number of synapses scales as the network size to the power of 3/2, the number of sequences P is proportional to N. In other words, sequential memory is scalable. Further-more, we find that there is an optimal ratio r between silent and nonsilent synapses at which the storage capacity $\alpha = P/[c\ (1 + r)N]$ assumes a maximum. For long sequences, the capacity of sequential memory is about one order of magnitude below the capacity for minimal sequences, but otherwise behaves similar to the case of minimal sequences. In a biologically inspired scenario, the information content per synapse is far below theoretical optimality, suggesting that the brain trades off error tolerance against information content in encoding sequential memories.

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## 61. Memory Capacity for Sequences in a Recurrent Network with Biological Constraints

*Christian Leibold, Richard Kempter*

https://doi.org/10.1162/089976606775774714 | pp. 904-941

The CA3 region of the hippocampus is a recurrent neural network that is essential for the storage and replay of sequences of patterns that represent behavioral events. Here we present a theoretical framework to calculate a sparsely connected network's capacity to store such sequences. As in CA3, only a limited subset of neurons in the network is active at any one time, pattern retrieval is subject to error, and the resources for plasticity are limited. Our analysis combines an analytical mean field approach, stochastic dynamics, and cellular simulations of a time-discrete McCulloch-Pitts network with binary synapses. To maximize the number of sequences that can be stored in the network, we concurrently optimize the number of active neurons, that is, pattern size, and the firing threshold. We find that for one-step associations (i.e., minimal sequences), the optimal pattern size is inversely proportional to the mean connectivity c, whereas the optimal firing threshold is independent of the connectivity. If the number of synapses per neuron is fixed, the maximum number P of stored sequences in a sufficiently large, nonmodular network is independent of its number N of cells. On the other hand, if the number of synapses scales as the network size to the power of 3/2, the number of sequences P is proportional to N. In other words, sequential memory is scalable. Furthermore, we find that there is an optimal ratio r between silent and nonsilent synapses at which the

storage capacity alpha = P//[c(1 + r)N] assumes a maximum. For long sequences, the capacity of sequential memory is about one order of magnitude below the capacity for minimal sequences, but otherwise behaves similar to the case of minimal sequences. In a biologically inspired scenario, the information content per synapse is far below theoretical optimality, suggesting that the brain trades off error tolerance against information content in encoding sequential memories.

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## 62. Kernel Fisher Discriminants for Outlier Detection

*Volker Röth*

https://doi.org/10.1162/neco.2006.18.4.942 | pp. 942-960

The problem of detecting atypical objects or outliers is one of the classical topics in (robust) statistics. Recently, it has been proposed to address this problem by means of one-class SVM classifiers. The method presented in this letter bridges the gap between kernelized one-class classification and gaussian density estimation in the induced feature space. Having established the exact relation between the two concepts, it is now possible to identify atypical objects by quantifying their deviations from the gaussian model. This model-based formalization of outliers overcomes the main conceptual shortcoming of most one-class approaches, which, in a strict sense, are unable to detect outliers, since the expected fraction of outliers has to be specified in advance. In order to overcome the inherent model selection problem of unsupervised kernel methods, a cross-validated likelihood criterion for selecting all free model parameters is applied. Experiments for detecting atypical objects in image databases effectively demonstrate the applicability of the proposed method in real-world scenarios.

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## 63. Kernel Fisher Discriminants for Outlier Detection

*Volker Röth*

https://doi.org/10.1162/089976606775774679 | pp. 942-960

The problem of detecting atypical objects or outliers is one of the classical topics in (robust) statistics. Recently, it has been proposed to address this problem by means of one-class SVM classifiers. The method presented in this letter bridges the gap between kernelized one-class classification and gaussian density estimation in the induced feature space. Having established the exact relation between the two concepts, it is now possible to identify atypical objects by quantifying their deviations from the gaussian model. This model-based formalization of outliers overcomes the main conceptual shortcoming of most one-class approaches, which, in a strict sense, are unable to detect outliers, since the expected fraction of outliers has to be specified in advance. In order to overcome the inherent model selection problem of unsupervised kernel methods, a cross-validated likelihood criterion for selecting all free model parameters is applied. Experiments for detecting atypical objects in image databases effectively demonstrate the applicability of the proposed method in real-world scenarios.

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## 64. Feature Scaling for Kernel Fisher Discriminant Analysis Using Leave-One-Out Cross Validation

*Liefeng Bo, Ling Wang, Licheng Jiao*

https://doi.org/10.1162/neco.2006.18.4.961 | pp. 961-978

Kernel fisher discriminant analysis (KFD) is a successful approach to classification. It is well known that the key challenge in KFD lies in the selection of free parameters such as kernel parameters and regularization parameters. Here we focus on the feature-scaling kernel where each feature individually associates with a scaling factor. A novel algorithm, named FS-KFD, is developed to tune the scaling factors and regularization parameters for the feature-scaling kernel. The proposed algorithm is based on optimizing the smooth leave-one-out error via a gradient-descent method and has been demonstrated to be computationally feasible. FS-KFD is motivated by the following two fundamental facts: the leave-one-out error of KFD can be expressed in closed form and the step function can be approximated by a sigmoid function. Empirical comparisons on artificial and benchmark data sets suggest that FS-KFD improves KFD in terms of classification accuracy.

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## 65. Feature Scaling for Kernel Fisher Discriminant Analysis Using Leave-One-Out Cross Validation

*Liefeng Bo, Ling Wang, Licheng Jiao*

https://doi.org/10.1162/089976606775774642 | pp. 961-978

Kernel fisher discriminant analysis (KFD) is a successful approach to classification. It is well known that the key challenge in KFD lies in the selection of free parameters such as kernel parameters and regularization parameters. Here we focus on the feature-scaling kernel where each feature individually associates with a scaling factor. A novel algorithm, named FS-KFD, is developed to tune the scaling factors and regularization parameters for the feature-scaling kernel. The proposed algorithm is based on optimizing the smooth leave-one-out error via a gradient-descent method and has been demonstrated to be computationally feasible. FS-KFD is motivated by the following two fundamental facts: the leave-one-out error of KFD can be expressed in closed form and the step function can be approximated by a sigmoid function. Empirical comparisons on artificial and benchmark data sets suggest that FS-KFD improves KFD in terms of classification accuracy.

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## 66. Class-Incremental Generalized Discriminant Analysis

*Wenming Zheng*

https://doi.org/10.1162/neco.2006.18.4.979 | pp. 979-1006

Generalized discriminant analysis (GDA) is the nonlinear extension of the classical linear discriminant analysis (LDA) via the kernel trick. Mathematically, GDA aims to solve a generalized eigenequation problem, which is always implemented by the use of singular value decomposition (SVD) in the previously proposed GDA algorithms. A major drawback of SVD, however, is the difficulty of designing an incremental solution for the eigenvalue problem. Moreover, there are still numerical problems of computing the eigenvalue problem of large matrices. In this article, we propose another algorithm for solving GDA as for the case of small sample size problem, which applies QR decomposition rather than SVD. A major contribution of the proposed algorithm is that it can incrementally update the discriminant vectors when new classes are inserted into the training set. The other major contribution of this article is the presentation of the modified kernel Gram-Schmidt (MKGS) orthogonalization algorithm for implementing the QR decomposition in the

feature space, which is more numerically stable than the kernel Gram-Schmidt (KGS) algorithm. We conduct experiments on both simulated and real data to demonstrate the better performance of the proposed methods.

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## 67. Class-Incremental Generalized Discriminant Analysis

*Wenming Zheng*

https://doi.org/10.1162/089976606775774633 | pp. 979-1006

Generalized discriminant analysis (GDA) is the nonlinear extension of the classical linear discriminant analysis (LDA) via the kernel trick. Mathematically, GDA aims to solve a generalized eigenequation problem, which is always implemented by the use of singular value decomposition (SVD) in the previously proposed GDA algorithms. A major drawback of SVD, however, is the difficulty of designing an incremental solution for the eigenvalue problem. Moreover, there are still numerical problems of computing the eigenvalue problem of large matrices. In this article, we propose another algorithm for solving GDA as for the case of small sample size problem, which applies QR decomposition rather than SVD. A major contribution of the proposed algorithm is that it can incrementally update the discriminant vectors when new classes are inserted into the training set. The other major contribution of this article is the presentation of the modified kernel Gram-Schmidt (MKGS) orthogonalization algorithm for implementing the QR decomposition in the feature space, which is more numerically stable than the kernel Gram-Schmidt (KGS) algorithm. We conduct experiments on both simulated and real data to demonstrate the better performance of the proposed methods.

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## 68. Singularities Affect Dynamics of Learning in Neuromanifolds

■■■-■■■ ■■■■■*, Hyeyoung Park, Tomoko Ozeki*

https://doi.org/10.1162/089976606776241002 | pp. 1007-1065

The parameter spaces of hierarchical systems such as multilayer perceptrons include singularities due to the symmetry and degeneration of hidden units. A parameter space forms a geometrical manifold, called the neuromanifold in the case of neural networks. Such a model is identified with a statistical model, and a Riemannian metric is given by the Fisher information matrix. However, the matrix degenerates at singularities. Such a singular structure is ubiquitous not only in multilayer perceptrons but also in the gaussian mixture probability densities, ARMA time-series model, and many other cases. The standard statistical paradigm of the Cramér-Rao theorem does not hold, and the singularity gives rise to strange behaviors in parameter estimation, hypothesis testing, Bayesian inference, model selection, and in particular, the dynamics of learning from examples. Prevailing theories so far have not paid much attention to the problem caused by singularity, relying only on ordinary statistical theories developed for regular (nonsingular) models. Only recently have researchers remarked on the effects of singularity, and theories are now being developed. This article gives an overview of the phenomena caused by the singularities of statistical manifolds related to multilayer perceptrons and gaussian mixtures. We demonstrate our recent results on these problems. Simple toy models are also used to show explicit solutions. We explain that the maximum likelihood estimator is no longer subject to the gaussian distribution even asymptotically, because the Fisher information matrix degenerates, that the model selection criteria such as AIC, BIC, and MDL fail to hold in these models, that a smooth Bayesian prior becomes singular in such models, and that the

trajectories of dynamics of learning are strongly affected by the singularity, causing plateaus or slow manifolds in the parameter space. The natural gradient method is shown to perform well because it takes the singular geometrical structure into account. The generalization error and the training error are studied in some examples.

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## 69. Singularities Affect Dynamics of Learning in Neuromanifolds

■■■-■■■ ■■■■■, *Hyeyoung Park, Tomoko Ozeki*

https://doi.org/10.1162/neco.2006.18.5.1007 | pp. 1007-1065

The parameter spaces of hierarchical systems such as multilayer perceptrons include singularities due to the symmetry and degeneration of hidden units. A parameter space forms a geometrical manifold, called the neuromanifold in the case of neural networks. Such a model is identified with a statistical model, and a Riemannian metric is given by the Fisher information matrix. However, the matrix degenerates at singularities. Such a singular structure is ubiquitous not only in multilayer perceptrons but also in the gaussian mixture probability densities, ARMA time-series model, and many other cases. The standard statistical paradigm of the Cramér-Rao theorem does not hold, and the singularity gives rise to strange behaviors in parameter estimation, hypothesis testing, Bayesian inference, model selection, and in particular, the dynamics of learning from examples. Prevailing theories so far have not paid much attention to the problem caused by singularity, relying only on ordinary statistical theories developed for regular (nonsingular) models. Only recently have researchers remarked on the effects of singularity, and theories are now being developed. This article gives an overview of the phenomena caused by the singularities of statistical manifolds related to multilayer perceptrons and gaussian mixtures. We demonstrate our recent results on these problems. Simple toy models are also used to show explicit solutions. We explain that the maximum likelihood estimator is no longer subject to the gaussian distribution even asymptotically, because the Fisher information matrix degenerates, that the model selection criteria such as AIC, BIC, and MDL fail to hold in these models, that a smooth Bayesian prior becomes singular in such models, and that the trajectories of dynamics of learning are strongly affected by the singularity, causing plateaus or slow manifolds in the parameter space. The natural gradient method is shown to perform well because it takes the singular geometrical structure into account. The generalization error and the training error are studied in some examples.

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## 70. How Noise Affects the Synchronization Properties of Recurrent Networks of Inhibitory Neurons

*Nicolas Brunel, D. Hansel*

https://doi.org/10.1162/neco.2006.18.5.1066 | pp. 1066-1110

GABAergic interneurons play a major role in the emergence of various types of synchronous oscillatory patterns of activity in the central nervous system. Motivated by these experimental facts, modeling studies have investigated mechanisms for the emergence of coherent activity in networks of inhibitory neurons. However, most of these studies have focused either when the noise in the network is absent or weak or in the opposite situation when it is strong. Hence, a full picture of how noise affects the dynamics of such systems is still lacking. The aim of this letter is to provide a more comprehensive understanding of the mechanisms by which the asynchronous states in large, fully connected networks of inhibitory neurons are

destabilized as a function of the noise level. Three types of single neuron models are considered: the leaky integrate-and-fire (LIF) model, the exponential integrate-and-fire (EIF), model and conductance-based models involving sodium and potassium Hodgkin-Huxley (HH) currents. We show that in all models, the instabilities of the asynchronous state can be classified in two classes. The first one consists of clustering instabilities, which exist in a restricted range of noise. These instabilities lead to synchronous patterns in which the population of neurons is broken into clusters of synchronously firing neurons. The irregularity of the firing patterns of the neurons is weak. The second class of instabilities, termed oscillatory firing rate instabilities, exists at any value of noise. They lead to cluster state at low noise. As the noise is increased, the instability occurs at larger coupling, and the pattern of firing that emerges becomes more irregular. In the regime of high noise and strong coupling, these instabilities lead to stochastic oscillations in which neurons fire in an approximately Poisson way with a common instantaneous probability of firing that oscillates in time.

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## 71. How Noise Affects the Synchronization Properties of Recurrent Networks of Inhibitory Neurons

*Nicolas Brunel, D. Hansel*

https://doi.org/10.1162/089976606776241048 | pp. 1066-1110

GABAergic interneurons play a major role in the emergence of various types of synchronous oscillatory patterns of activity in the central nervous system. Motivated by these experimental facts, modeling studies have investigated mechanisms for the emergence of coherent activity in networks of inhibitory neurons. However, most of these studies have focused either when the noise in the network is absent or weak or in the opposite situation when it is strong. Hence, a full picture of how noise affects the dynamics of such systems is still lacking. The aim of this letter is to provide a more comprehensive understanding of the mechanisms by which the asynchronous states in large, fully connected networks of inhibitory neurons are destabilized as a function of the noise level. Three types of single neuron models are considered: the leaky integrate-and-fire (LIF) model, the exponential integrate-and-fire (EIF), model and conductance-based models involving sodium and potassium Hodgkin-Huxley (HH) currents. We show that in all models, the instabilities of the asynchronous state can be classified in two classes. The first one consists of clustering instabilities, which exist in a restricted range of noise. These instabilities lead to synchronous patterns in which the population of neurons is broken into clusters of synchronously firing neurons. The irregularity of the firing patterns of the neurons is weak. The second class of instabilities, termed oscillatory firing rate instabilities, exists at any value of noise. They lead to cluster state at low noise. As the noise is increased, the instability occurs at larger coupling, and the pattern of firing that emerges becomes more irregular. In the regime of high noise and strong coupling, these instabilities lead to stochastic oscillations in which neurons fire in an approximately Poisson way with a common instantaneous probability of firing that oscillates in time.

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## 72. Analysis of Synchronization Between Two Modules of Pulse Neural Networks with Excitatory and Inhibitory Connections

*Takashi Kanamaru*

https://doi.org/10.1162/neco.2006.18.5.1111 | pp. 1111-1131

To study the synchronized oscillations among distant neurons in the visual cortex, we analyzed the synchronization between two modules of pulse neural networks using the phase response function. It was

found that the intermodule connections from excitatory to excitatory ensembles tend to stabilize the antiphase synchronization and that the intermodule connections from excitatory to inhibitory ensembles tend to stabilize the in-phase synchronization. It was also found that the intermodule synchronization was more noticeable when the inner-module synchronization was weak.

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## 73. Analysis of Synchronization Between Two Modules of Pulse Neural Networks with Excitatory and Inhibitory Connections

*Takashi Kanamaru*

https://doi.org/10.1162/089976606776241039 | pp. 1111-1131

To study the synchronized oscillations among distant neurons in the visual cortex, we analyzed the synchronization between two modules of pulse neural networks using the phase response function. It was found that the intermodule connections from excitatory to excitatory ensembles tend to stabilize the antiphase synchronization and that the intermodule connections from excitatory to inhibitory ensembles tend to stabilize the in-phase synchronization. It was also found that the intermodule synchronization was more noticeable when the inner-module synchronization was weak.

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## 74. A Sensorimotor Map: Modulating Lateral Interactions for Anticipation and Planning

*Marc Toussaint*

https://doi.org/10.1162/089976606776240995 | pp. 1132-1155

Experimental studies of reasoning and planned behavior have provided evidence that nervous systems use internal models to perform predictive motor control, imagery, inference, and planning. Classical (model-free) reinforcement learning approaches omit such a model; standard sensorimotor models account for forward and backward functions of sensorimotor dependencies but do not provide a proper neural representation on which to realize planning. We propose a sensorimotor map to represent such an internal model. The map learns a state representation similar to self-organizing maps but is inherently coupled to sensor and motor signals. Motor activations modulate the lateral connection strengths and thereby induce anticipatory shifts of the activity peak on the sensorimotor map. This mechanism encodes a model of the change of stimuli depending on the current motor activities. The activation dynamics on the map are derived from neural field models. An additional dynamic process on the sensorimotor map (derived from dynamic programming) realizes planning and emits corresponding goal-directed motor sequences, for instance, to navigate through a maze.

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## 75. A Sensorimotor Map: Modulating Lateral Interactions for Anticipation and Planning

*Marc Toussaint*

https://doi.org/10.1162/neco.2006.18.5.1132 | pp. 1132-1155

Experimental studies of reasoning and planned behavior have provided evidence that nervous systems use internal models to perform predictive motor control, imagery, inference, and planning. Classical (model-free)

reinforcement learning approaches omit such a model; standard sensorimotor models account for forward and backward functions of sensorimotor dependencies but do not provide a proper neural representation on which to realize planning. We propose a sensorimotor map to represent such an internal model. The map learns a state representation similar to self-organizing maps but is inherently coupled to sensor and motor signals. Motor activations modulate the lateral connection strengths and thereby induce anticipatory shifts of the activity peak on the sensorimotor map. This mechanism encodes a model of the change of stimuli depending on the current motor activities. The activation dynamics on the map are derived from neural field models. An additional dynamic process on the sensorimotor map (derived from dynamic programming) realizes planning and emits corresponding goal-directed motor sequences, for instance, to navigate through a maze.

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## 76. A Reflexive Neural Network for Dynamic Biped Walking Control

*Tao Geng, Bernd Porr, Florentin Wörgötter*

https://doi.org/10.1162/neco.2006.18.5.1156 | pp. 1156-1196

Biped walking remains a difficult problem, and robot models can greatly facilitate our understanding of the underlying biomechanical principles as well as their neuronal control. The goal of this study is to specifically demonstrate that stable biped walking can be achieved by combining the physical properties of the walking robot with a small, reflex-based neuronal network governed mainly by local sensor signals. Building on earlier work (Taga, 1995; Cruse, Kindermann, Schumm, Dean, & Schmitz, 1998), this study shows that human-like gaits emerge without specific position or trajectory control and that the walker is able to compensate small disturbances through its own dynamical properties. The reflexive controller used here has the following characteristics, which are different from earlier approaches: (1) Control is mainly local. Hence, it uses only two signals (anterior extreme angle and ground contact), which operate at the interjoint level. All other signals operate only at single joints. (2) Neither position control nor trajectory tracking control is used. Instead, the approximate nature of the local reflexes on each joint allows the robot mechanics itself (e.g., its passive dynamics) to contribute substantially to the overall gait trajectory computation. (3) The motor control scheme used in the local reflexes of our robot is more straightforward and has more biological plausibility than that of other robots, because the outputs of the motor neurons in our reflexive controller are directly driving the motors of the joints rather than working as references for position or velocity control. As a consequence, the neural controller and the robot mechanics are closely coupled as a neuromechanical system, and this study emphasizes that dynamically stable biped walking gaits emerge from the coupling between neural computation and physical computation. This is demonstrated by different walking experiments using a real robot as well as by a Poincaré map analysis applied on a model of the robot in order to assess its stability.

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## 77. A Reflexive Neural Network for Dynamic Biped Walking Control

*Tao Geng, Bernd Porr, Florentin Wörgötter*

https://doi.org/10.1162/089976606776241057 | pp. 1156-1196

Biped walking remains a difficult problem, and robot models can greatly facilitate our understanding of the underlying biomechanical principles as well as their neuronal control. The goal of this study is to specifically demonstrate that stable biped walking can be achieved by combining the physical properties of the walking

robot with a small, reflex-based neuronal network governed mainly by local sensor signals. Building on earlier work (Taga, 1995; Cruse, Kindermann, Schumm, Dean, & Schmitz, 1998), this study shows that human-like gaits emerge without specific position or trajectory control and that the walker is able to compensate small disturbances through its own dynamical properties. The reflexive controller used here has the following characteristics, which are different from earlier approaches: (1) Control is mainly local. Hence, it uses only two signals (anterior extreme angle and ground contact), which operate at the interjoint level. All other signals operate only at single joints. (2) Neither position control nor trajectory tracking control is used. Instead, the approximate nature of the local reflexes on each joint allows the robot mechanics itself (e.g., its passive dynamics) to contribute substantially to the overall gait trajectory computation. (3) The motor control scheme used in the local reflexes of our robot is more straightforward and has more biological plausibility than that of other robots, because the outputs of the motor neurons in our reflexive controller are directly driving the motors of the joints rather than working as references for position or velocity control. As a consequence, the neural controller and the robot mechanics are closely coupled as a neuromechanical system, and this study emphasizes that dynamically stable biped walking gaits emerge from the coupling between neural computation and physical computation. This is demonstrated by different walking experiments using a real robot as well as by a Poincaré map analysis applied on a model of the robot in order to assess its stability.

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## 78. A Set Probability Technique for Detecting Relative Time Order Across Multiple Neurons

*Anne C. Smith, Peter Smith*

https://doi.org/10.1162/089976606776241020 | pp. 1197-1214

With the development of multielectrode recording techniques, it is possible to measure the cell firing patterns of multiple neurons simultaneously, generating a large quantity of data. Identification of the firing patterns within these large groups of cells is an important and a challenging problem in data analysis. Here, we consider the problem of measuring the significance of a repeat in the cell firing sequence across arbitrary numbers of cells. In particular, we consider the question, given a ranked order of cells numbered 1 to N, what is the probability that another sequence of length n contains j consecutive increasing elements? Assuming each element of the sequence is drawn with replacement from the numbers 1 through N, we derive a recursive formula for the probability of the sequence of length j or more. For n<2j, a closed-form solution is derived. For n>or=2j, we obtain upper and lower bounds for these probabilities for various combinations of parameter values. These can be computed very quickly. For a typical case with small N (<10) and large n (<3000), sequences of 7 and 8 are statistically very unlikely. A potential application of this technique is in the detection of repeats in hippocampal place cell order during sleep. Unlike most previous articles on increasing runs in random lists, we use a probability approach based on sets of overlapping sequences.

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## 79. A Set Probability Technique for Detecting Relative Time Order Across Multiple Neurons

*Anne C. Smith, Peter Smith*

https://doi.org/10.1162/neco.2006.18.5.1197 | pp. 1197-1214

With the development of multielectrode recording techniques, it is possible to measure the cell firing patterns of multiple neurons simultaneously, generating a large quantity of data. Identification of the firing patterns within these large groups of cells is an important and a challenging problem in data analysis. Here, we consider the problem of measuring the significance of a repeat in the cell firing sequence across arbitrary numbers of cells. In particular, we consider the question, given a ranked order of cells numbered 1 to N, what is the probability that another sequence of length n contains j consecutive increasing elements? Assuming each element of the sequence is drawn with replacement from the numbers 1 through N, we derive a recursive formula for the probability of the sequence of length j or more. For n < 2j, a closed-form solution is derived. For n ≥ 2j, we obtain upper and lower bounds for these probabilities for various combinations of parameter values. These can be computed very quickly. For a typical case with small N (<10) and large n (<3000), sequences of 7 and 8 are statistically very unlikely. A potential application of this technique is in the detection of repeats in hippocampal place cell order during sleep. Unlike most previous articles on increasing runs in random lists, we use a probability approach based on sets of overlapping sequences.

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## 80. Payoff-Monotonic Game Dynamics and the Maximum Clique Problem

*Marcello Pelillo, Andrea Torsello*

https://doi.org/10.1162/neco.2006.18.5.1215 | pp. 1215-1258

Evolutionary game-theoretic models and, in particular, the so-called replicator equations have recently proven to be remarkably effective at approximately solving the maximum clique and related problems. The approach is centered around a classic result from graph theory that formulates the maximum clique problem as a standard (continuous) quadratic program and exploits the dynamical properties of these models, which, under a certain symmetry assumption, possess a Lyapunov function. In this letter, we generalize previous work along these lines in several respects. We introduce a wide family of game-dynamic equations known as payoff-monotonic dynamics, of which replicator dynamics are a special instance, and show that they enjoy precisely the same dynamical properties as standard replicator equations. These properties make any member of this family a potential heuristic for solving standard quadratic programs and, in particular, the maximum clique problem. Extensive simulations, performed on random as well as DIMACS benchmark graphs, show that this class contains dynamics that are considerably faster than and at least as accurate as replicator equations. One problem associated with these models, however, relates to their inability to escape from poor local solutions. To overcome this drawback, we focus on a particular subclass of payoff-monotonic dynamics used to model the evolution of behavior via imitation processes and study the stability of their equilibria when a regularization parameter is allowed to take on negative values. A detailed analysis of these properties suggests a whole class of annealed imitation heuristics for the maximum clique problem, which are based on the idea of varying the parameter during the imitation optimization process in a principled way, so as to avoid unwanted inefficient solutions. Experiments show that the proposed annealing procedure does help to avoid poor local optima by initially driving the dynamics toward promising regions in state space. Furthermore, the models outperform state-of-the-art neural network algorithms for maximum clique, such as mean field annealing, and compare well with powerful continuous-based heuristics.

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## 81. Payoff-Monotonic Game Dynamics and the Maximum Clique Problem

*Marcello Pelillo, Andrea Torsello*

https://doi.org/10.1162/089976606776241011 | pp. 1215-1258

Evolutionary game-theoretic models and, in particular, the so-called replicator equations have recently proven to be remarkably effective at approximately solving the maximum clique and related problems. The approach is centered around a classic result from graph theory that formulates the maximum clique problem as a standard (continuous) quadratic program and exploits the dynamical properties of these models, which, under a certain symmetry assumption, possess a Lyapunov function. In this letter, we generalize previous work along these lines in several respects. We introduce a wide family of game-dynamic equations known as payoff-monotonic dynamics, of which replicator dynamics are a special instance, and show that they enjoy precisely the same dynamical properties as standard replicator equations. These properties make any member of this family a potential heuristic for solving standard quadratic programs and, in particular, the maximum clique problem. Extensive simulations, performed on random as well as DIMACS benchmark graphs, show that this class contains dynamics that are considerably faster than and at least as accurate as replicator equations. One problem associated with these models, however, relates to their inability to escape from poor local solutions. To overcome this drawback, we focus on a particular subclass of payoff-monotonic dynamics used to model the evolution of behavior via imitation processes and study the stability of their equilibria when a regularization parameter is allowed to take on negative values. A detailed analysis of these properties suggests a whole class of annealed imitation heuristics for the maximum clique problem, which are based on the idea of varying the parameter during the imitation optimization process in a principled way, so as to avoid unwanted inefficient solutions. Experiments show that the proposed annealing procedure does help to avoid poor local optima by initially driving the dynamics toward promising regions in state space. Furthermore, the models outperform state-of-the-art neural network algorithms for maximum clique, such as mean field annealing, and compare well with powerful continuous-based heuristics.

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## 82. Correlation and Independence in the Neural Code

■■■-■■■ ■■■■■, *Hiroyuki Nakahara*

https://doi.org/10.1162/neco.2006.18.6.1259 | pp. 1259-1267

The decoding scheme of a stimulus can be different from the stochastic encoding scheme in the neural population coding. The stochastic fluctuations are not independent in general, but an independent version could be used for the ease of decoding. How much information is lost by using this unfaithful model for decoding? There are discussions concerning loss of information (Nirenberg & Latham, 2003; Schneidman, Bialek, & Berry, 2003). We elucidate the Nirenberg-Latham loss from the point of view of information geometry.

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## 83. Analysis of Spike Statistics in Neuronal Systems with Continuous Attractors or Multiple, Discrete Attractor States

*Paul Miller*

https://doi.org/10.1162/neco.2006.18.6.1268 | pp. 1268-1317

Attractor networks are likely to underlie working memory and integrator circuits in the brain. It is unknown whether continuous quantities are stored in an analog manner or discretized and stored in a set of discrete

attractors. In order to investigate the important issue of how to differentiate the two systems, here we compare the neuronal spiking activity that arises from a continuous (line) attractor with that from a series of discrete attractors. Stochastic fluctuations cause the position of the system along its continuous attractor to vary as a random walk, whereas in a discrete attractor, noise causes spontaneous transitions to occur between discrete states at random intervals. We calculate the statistics of spike trains of neurons firing as a Poisson process with rates that vary according to the underlying attractor network. Since individual neurons fire spikes probabilistically and since the state of the network as a whole drifts randomly, the spike trains of individual neurons follow a doubly stochastic (Poisson) point process. We compare the series of spike trains from the two systems using the autocorrelation function, Fano factor, and interspike interval (ISI) distribution. Although the variation in rate can be dramatically different, especially for short time intervals, surprisingly both the autocorrelation functions and Fano factors are identical, given appropriate scaling of the noise terms. Since the range of firing rates is limited in neurons, we also investigate systems for which the variation in rate is bounded by either rigid limits or because of leak to a single attractor state, such as the Ornstein-Uhlenbeck process. In these cases, the time dependence of the variance in rate can be different between discrete and continuous systems, so that in principle, these processes can be distinguished using second-order spike statistics.

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## 84. Optimal Spike-Timing-Dependent Plasticity for Precise Action Potential Firing in Supervised Learning

*Jean-Pascal Pfister, Taro Toyoizumi, David Barber, Wulfram Gerstner*

https://doi.org/10.1162/neco.2006.18.6.1318 | pp. 1318-1348

In timing-based neural codes, neurons have to emit action potentials at precise moments in time. We use a supervised learning paradigm to derive a synaptic update rule that optimizes by gradient ascent the likelihood of postsynaptic firing at one or several desired firing times. We find that the optimal strategy of up- and downregulating synaptic efficacies depends on the relative timing between presynaptic spike arrival and desired postsynaptic firing. If the presynaptic spike arrives before the desired postsynaptic spike timing, our optimal learning rule predicts that the synapse should become potentiated. The dependence of the potentiation on spike timing directly reflects the time course of an excitatory postsynaptic potential. However, our approach gives no unique reason for synaptic depression under reversed spike timing. In fact, the presence and amplitude of depression of synaptic efficacies for reversed spike timing depend on how constraints are implemented in the optimization problem. Two different constraints, control of postsynaptic rates and control of temporal locality, are studied. The relation of our results to spike-timing-dependent plasticity and reinforcement learning is discussed.

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## 85. When Response Variability Increases Neural Network Robustness to Synaptic Noise

*Gleb Basalyga, Emilio Salinas*

https://doi.org/10.1162/neco.2006.18.6.1349 | pp. 1349-1379

Cortical sensory neurons are known to be highly variable, in the sense that responses evoked by identical stimuli often change dramatically from trial to trial. The origin of this variability is uncertain, but it is usually interpreted as detrimental noise that reduces the computational accuracy of neural circuits. Here we

investigate the possibility that such response variability might in fact be beneficial, because it may partially compensate for a decrease in accuracy due to stochastic changes in the synaptic strengths of a network. We study the interplay between two kinds of noise, response (or neuronal) noise and synaptic noise, by analyzing their joint influence on the accuracy of neural networks trained to perform various tasks. We find an interesting, generic interaction: when fluctuations in the synaptic connections are proportional to their strengths (multiplicative noise), a certain amount of response noise in the input neurons can significantly improve network performance, compared to the same network without response noise. Performance is enhanced because response noise and multiplicative synaptic noise are in some ways equivalent. So if the algorithm used to find the optimal synaptic weights can take into account the variability of the model neurons, it can also take into account the variability of the synapses. Thus, the connection patterns generated with response noise are typically more resistant to synaptic degradation than those obtained without response noise. As a consequence of this interplay, if multiplicative synaptic noise is present, it is better to have response noise in the network than not to have it. These results are demonstrated analytically for the most basic network consisting of two input neurons and one output neuron performing a simple classification task, but computer simulations show that the phenomenon persists in a wide range of architectures, including recurrent (attractor) networks and sensorimotor networks that perform coordinate transformations. The results suggest that response variability could play an important dynamic role in networks that continuously learn.

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## 86. Strongly Improved Stability and Faster Convergence of Temporal Sequence Learning by Using Input Correlations Only

*Bernd Porr, Florentin Wörgötter*

https://doi.org/10.1162/neco.2006.18.6.1380 | pp. 1380-1412

Currently all important, low-level, unsupervised network learning algorithms follow the paradigm of Hebb, where input and output activity are correlated to change the connection strength of a synapse. However, as a consequence, classical Hebbian learning always carries a potentially destabilizing autocorrelation term, which is due to the fact that every input is in a weighted form reflected in the neuron's output. This self-correlation can lead to positive feedback, where increasing weights will increase the output, and vice versa, which may result in divergence. This can be avoided by different strategies like weight normalization or weight saturation, which, however, can cause different problems. Consequently, in most cases, high learning rates cannot be used for Hebbian learning, leading to relatively slow convergence. Here we introduce a novel correlation-based learning rule that is related to our isotropic sequence order (ISO) learning rule (Porr & Wörgötter, 2003a), but replaces the derivative of the output in the learning rule with the derivative of the reflex input. Hence, the new rule uses input correlations only, effectively implementing strict heterosynaptic learning. This looks like a minor modification but leads to dramatically improved properties. Elimination of the output from the learning rule removes the unwanted, destabilizing autocorrelation term, allowing us to use high learning rates. As a consequence, we can mathematically show that the theoretical optimum of one-shot learning can be reached under ideal conditions with the new rule. This result is then tested against four different experimental setups, and we will show that in all of them, very few (and sometimes only one) learning experiences are needed to achieve the learning goal. As a consequence, the new learning rule is up to 100 times faster and in general more stable than ISO learning.

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## 87. An Oscillatory Neural Model of Multiple Object Tracking

*Yakov Kazanovich, Roman Borisyuk*

https://doi.org/10.1162/neco.2006.18.6.1413 | pp. 1413-1440

An oscillatory neural network model of multiple object tracking is described. The model works with a set of identical visual objects moving around the screen. At the initial stage, the model selects into the focus of attention a subset of objects initially marked as targets. Other objects are used as distractors. The model aims to preserve the initial separation between targets and distractors while objects are moving. This is achieved by a proper interplay of synchronizing and desynchronizing interactions in a multilayer network, where each layer is responsible for tracking a single target. The results of the model simulation are presented and compared with experimental data. In agreement with experimental evidence, simulations with a larger number of targets have shown higher error rates. Also, the functioning of the model in the case of temporarily overlapping objects is presented.

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## 88. Analysis of Cluttered Scenes Using an Elastic Matching Approach for Stereo Images

*Christian Eckes, Jochen Triesch, Christoph von der Malsburg*

https://doi.org/10.1162/neco.2006.18.6.1441 | pp. 1441-1471

We present a system for the automatic interpretation of cluttered scenes containing multiple partly occluded objects in front of unknown, complex backgrounds. The system is based on an extended elastic graph matching algorithm that allows the explicit modeling of partial occlusions. Our approach extends an earlier system in two ways. First, we use elastic graph matching in stereo image pairs to increase matching robustness and disambiguate occlusion relations. Second, we use richer feature descriptions in the object models by integrating shape and texture with color features. We demonstrate that the combination of both extensions substantially increases recognition performance. The system learns about new objects in a simple one-shot learning approach. Despite the lack of statistical information in the object models and the lack of an explicit background model, our system performs surprisingly well for this very difficult task. Our results underscore the advantages of view-based feature constellation representations for difficult object recognition problems.

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## 89. Support Vector Machines for Dyadic Data

*Sepp Hochreiter, Klaus Obermayer*

https://doi.org/10.1162/neco.2006.18.6.1472 | pp. 1472-1510

We describe a new technique for the analysis of dyadic data, where two sets of objects (row and column objects) are characterized by a matrix of numerical values that describe their mutual relationships. The new technique, called potential support vector machine (P-SVM), is a large-margin method for the construction of classifiers and regression functions for the column objects. Contrary to standard support vector machine approaches, the P-SVM minimizes a scale-invariant capacity measure and requires a new set of constraints. As a result, the P-SVM method leads to a usually sparse expansion of the classification and regression functions in terms of the row rather than the column objects and can handle data and kernel matrices that are neither positive definite nor square. We then describe two complementary regularization schemes. The

first scheme improves generalization performance for classification and regression tasks; the second scheme leads to the selection of a small, informative set of row support objects and can be applied to feature selection. Benchmarks for classification, regression, and feature selection tasks are performed with toy data as well as with several real-world data sets. The results show that the new method is at least competitive with but often performs better than the benchmarked standard methods for standard vectorial as well as true dyadic data sets. In addition, a theoretical justification is provided for the new approach.

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## 90. Optimal Neuronal Tuning for Finite Stimulus Spaces

*William M. Brown, Alex Bäcker*

https://doi.org/10.1162/neco.2006.18.7.1511 | pp. 1511-1526

The efficiency of neuronal encoding in sensory and motor systems has been proposed as a first principle governing response properties within the central nervous system. We present a continuation of a theoretical study presented by Zhang and Sejnowski, where the influence of neuronal tuning properties on encoding accuracy is analyzed using information theory. When a finite stimulus space is considered, we show that the encoding accuracy improves with narrow tuning for one- and two-dimensional stimuli. For three dimensions and higher, there is an optimal tuning width.

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## 91. A Fast Learning Algorithm for Deep Belief Nets

*Geoffrey E. Hinton, Simon Osindero, Yee■Whye Teh*

https://doi.org/10.1162/neco.2006.18.7.1527 | pp. 1527-1554

We show how to use "complementary priors" to eliminate the explaining-away effects that make inference difficult in densely connected belief nets that have many hidden layers. Using complementary priors, we derive a fast, greedy algorithm that can learn deep, directed belief networks one layer at a time, provided the top two layers form an undirected associative memory. The fast, greedy algorithm is used to initialize a slower learning procedure that fine-tunes the weights using a contrastive version of the wake-sleep algorithm. After fine-tuning, a network with three hidden layers forms a very good generative model of the joint distribution of handwritten digit images and their labels. This generative model gives better digit classification than the best discriminative learning algorithms. The low-dimensional manifolds on which the digits lie are modeled by long ravines in the free-energy landscape of the top-level associative memory, and it is easy to explore these ravines by using the directed connections to display what the associative memory has in mind.

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## 92. Optimal Tuning Widths in Population Coding of Periodic Variables

*Marcelo A. Montemurro, Stefano Panzeri*

https://doi.org/10.1162/neco.2006.18.7.1555 | pp. 1555-1576

We study the relationship between the accuracy of a large neuronal population in encoding periodic sensory stimuli and the width of the tuning curves of individual neurons in the population. By using general simple models of population activity, we show that when considering one or two periodic stimulus features, a narrow tuning width provides better population encoding accuracy. When encoding more than two periodic

stimulus features, the information conveyed by the population is instead maximal for finite values of the tuning width. These optimal values are only weakly dependent on model parameters and are similar to the width of tuning to orientation or motion direction of real visual cortical neurons. A very large tuning width leads to poor encoding accuracy, whatever the number of stimulus features encoded. Thus, optimal coding of periodic stimuli is different from that of nonperiodic stimuli, which, as shown in previous studies, would require infinitely large tuning widths when coding more than two stimulus features.

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## 93. How Inhibitory Oscillations Can Train Neural Networks and Punish Competitors

*Kenneth A. Norman, Ehren L. Newman, Greg Detre, Sean M. Polyn*

https://doi.org/10.1162/neco.2006.18.7.1577 | pp. 1577-1610

We present a new learning algorithm that leverages oscillations in the strength of neural inhibition to train neural networks. Raising inhibition can be used to identify weak parts of target memories, which are then strengthened. Conversely, lowering inhibition can be used to identify competitors, which are then weakened. To update weights, we apply the Contrastive Hebbian Learning equation to successive time steps of the network. The sign of the weight change equation varies as a function of the phase of the inhibitory oscillation. We show that the learning algorithm can memorize large numbers of correlated input patterns without collapsing and that it shows good generalization to test patterns that do not exactly match studied patterns.

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## 94. Temporal Decoding by Phase-Locked Loops: Unique Features of Circuit-Level Implementations and Their Significance for Vibrissal Information Processing

*Miriam Zacksenhouse, Ehud Ahissar*

https://doi.org/10.1162/neco.2006.18.7.1611 | pp. 1611-1636

Rhythmic active touch, such as whisking, evokes a periodic reference spike train along which the timing of a novel stimulus, induced, for example, when the whiskers hit an external object, can be interpreted. Previous work supports the hypothesis that the whisking-induced spike train entrains a neural implementation of a phase-locked loop (NPLL) in the vibrissal system. Here we extend this work and explore how the entrained NPLL decodes the delay of the novel, contact-induced stimulus and facilitates object localization. We consider two implementations of NPLLs, which are based on a single neuron or a neural circuit, respectively, and evaluate the resulting temporal decoding capabilities. Depending on the structure of the NPLL, it can lock in either a phase- or co-phase-sensitive mode, which is sensitive to the timing of the input with respect to the beginning of either the current or the next cycle, respectively. The co-phase-sensitive mode is shown to be unique to circuit-based NPLLs. Concentrating on temporal decoding in the vibrissal system of rats, we conclude that both the nature of the information processing task and the response characteristics suggest that the computation is sensitive to the co-phase. Consequently, we suggest that the underlying thalamocortical loop should implement a circuit-based NPLL.

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## 95. Representation and Timing in Theories of the Dopamine System

*Nathaniel D. Daw, Aaron Courville, David S. Touretzky*

https://doi.org/10.1162/neco.2006.18.7.1637 | pp. 1637-1677

Although the responses of dopamine neurons in the primate midbrain are well characterized as carrying a temporal difference (TD) error signal for reward prediction, existing theories do not offer a credible account of how the brain keeps track of past sensory events that may be relevant to predicting future reward. Empirically, these shortcomings of previous theories are particularly evident in their account of experiments in which animals were exposed to variation in the timing of events. The original theories mispredicted the results of such experiments due to their use of a representational device called a tapped delay line. Here we propose that a richer understanding of history representation and a better account of these experiments can be given by considering TD algorithms for a formal setting that incorporates two features not originally considered in theories of the dopaminergic response: partial observability (a distinction between the animal's sensory experience and the true underlying state of the world) and semi-Markov dynamics (an explicit account of variation in the intervals between events). The new theory situates the dopaminergic system in a richer functional and anatomical context, since it assumes (in accord with recent computational theories of cortex) that problems of partial observability and stimulus history are solved in sensory cortex using statistical modeling and inference and that the TD system predicts reward using the results of this inference rather than raw sensory data. It also accounts for a range of experimental data, including the experiments involving programmed temporal variability and other previously unmodeled dopaminergic response phenomena, which we suggest are related to subjective noise in animals' interval timing. Finally, it offers new experimental predictions and a rich theoretical framework for designing future experiments.

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## 96. Experiments with AdaBoost.RT, an Improved Boosting Scheme for Regression

*Durga Lal Shrestha, Dimitri Solomatine*

https://doi.org/10.1162/neco.2006.18.7.1678 | pp. 1678-1710

The application of boosting technique to regression problems has received relatively little attention in contrast to research aimed at classification problems. This letter describes a new boosting algorithm, AdaBoost.RT, for regression problems. Its idea is in filtering out the examples with the relative estimation error that is higher than the preset threshold value, and then following the AdaBoost procedure. Thus, it requires selecting the suboptimal value of the error threshold to demarcate examples as poorly or well predicted. Some experimental results using the M5 model tree as a weak learning machine for several benchmark data sets are reported. The results are compared to other boosting methods, bagging, artificial neural networks, and a single M5 model tree. The preliminary empirical comparisons show higher performance of AdaBoost.RT for most of the considered data sets.

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## 97. A Connectionist Computational Model for Epistemic and Temporal Reasoning

*Artur d'Avila Garcez, Luís C. Lamb*

https://doi.org/10.1162/neco.2006.18.7.1711 | pp. 1711-1738

The importance of the efforts to bridge the gap between the connectionist and symbolic paradigms of artificial intelligence has been widely recognized. The merging of theory (background knowledge) and data learning (learning from examples) into neural-symbolic systems has indicated that such a learning system is more effective than purely symbolic or purely connectionist systems. Until recently, however, neural-symbolic systems were not able to fully represent, reason, and learn expressive languages other than classical propositional and fragments of first-order logic. In this article, we show that nonclassical logics, in particular propositional temporal logic and combinations of temporal and epistemic (modal) reasoning, can be effectively computed by artificial neural networks. We present the language of a connectionist temporal logic of knowledge (CTLK). We then present a temporal algorithm that translates CTLK theories into ensembles of neural networks and prove that the translation is correct. Finally, we apply CTLK to the muddy children puzzle, which has been widely used as a test-bed for distributed knowledge representation. We provide a complete solution to the puzzle with the use of simple neural networks, capable of reasoning about knowledge evolution in time and of knowledge acquisition through learning.

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## 98. Multivariate Information Bottleneck

*Noam Slonim, Nir Friedman, Naftali Tishby*

https://doi.org/10.1162/neco.2006.18.8.1739 | pp. 1739-1789

The information bottleneck (IB) method is an unsupervised model independent data organization technique. Given a joint distribution, p(X, Y), this method constructs a new variable, T, that extracts partitions, or clusters, over the values of X that are informative about Y. Algorithms that are motivated by the IB method have already been applied to text classification, gene expression, neural code, and spectral analysis. Here, we introduce a general principled framework for multivariate extensions of the IB method. This allows us to consider multiple systems of data partitions that are interrelated. Our approach utilizes Bayesian networks for specifying the systems of clusters and which information terms should be maintained. We show that this construction provides insights about bottleneck variations and enables us to characterize the solutions of these variations. We also present four different algorithmic approaches that allow us to construct solutions in practice and apply them to several real-world problems.

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## 99. Variational Bayesian Multinomial Probit Regression with Gaussian Process Priors

*Mark Girolami, Simon Rogers*

https://doi.org/10.1162/neco.2006.18.8.1790 | pp. 1790-1817

It is well known in the statistics literature that augmenting binary and polychotomous response models with gaussian latent variables enables exact Bayesian analysis via Gibbs sampling from the parameter posterior. By adopting such a data augmentation strategy, dispensing with priors over regression coefficients in favor of gaussian process (GP) priors over functions, and employing variational approximations to the full posterior, we obtain efficient computational methods for GP classification in the multiclass setting. 1 The model augmentation with additional latent variables ensures full a posteriori class coupling while retaining the simple a priori independent GP covariance structure from which sparse approximations, such as multiclass informative vector machines (IVM), emerge in a natural and straightforward manner. This is the first time that a fully variational Bayesian treatment for multiclass GP classification has been developed

without having to resort to additional explicit approximations to the nongaussian likelihood term. Empirical comparisons with exact analysis use Markov Chain Monte Carlo (MCMC) and Laplace approximations illustrate the utility of the variational approximation as a computationally economic alternative to full MCMC and it is shown to be more accurate than the Laplace approximation.

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## 100. A Novel Neural Network for a Class of Convex Quadratic Minimax Problems

*Xingbao Gao, Li■Zhi Liao*

https://doi.org/10.1162/neco.2006.18.8.1818 | pp. 1818-1846

Based on the inherent properties of convex quadratic minimax problems, this article presents a new neural network model for a class of convex quadratic minimax problems. We show that the new model is stable in the sense of Lyapunov and will converge to an exact saddle point in finite time by defining a proper convex energy function. Furthermore, global exponential stability of the new model is shown under mild conditions. Compared with the existing neural networks for the convex quadratic minimax problem, the proposed neural network has finite-time convergence, a simpler structure, and lower complexity. Thus, the proposed neural network is more suitable for parallel implementation by using simple hardware units. The validity and transient behavior of the proposed neural network are illustrated by some simulation results.

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## 101. Dynamic Gain Changes During Attentional Modulation

*Arun P. Sripati, Kenneth O. Johnson*

https://doi.org/10.1162/neco.2006.18.8.1847 | pp. 1847-1867

Attention causes a multiplicative effect on firing rates of cortical neurons without affecting their selectivity (Motter, 1993; McAdams & Maunsell, 1999a) or the relationship between the spike count mean and variance (McAdams & Maunsell, 1999b). We analyzed attentional modulation of the firing rates of 144 neurons in the secondary somatosensory cortex (SII) of two monkeys trained to switch their attention between a tactile pattern recognition task and a visual task. We found that neurons in SII cortex also undergo a predominantly multiplicative modulation in firing rates without affecting the ratio of variance to mean firing rate (i.e., the Fano factor). Furthermore, both additive and multiplicative components of attentional modulation varied dynamically during the stimulus presentation. We then used a standard conductance-based integrate-and-fire model neuron to ascertain which mechanisms might account for a multiplicative increase in firing rate without affecting the Fano factor. Six mechanisms were identified as biophysically plausible ways that attention could modify the firing rate: spike threshold, firing rate adaptation, excitatory input synchrony, synchrony between all inputs, membrane leak resistance, and reset potential. Of these, only a change in spike threshold or in firing rate adaptation affected model firing rates in a manner compatible with the observed neural data. The results indicate that only a limited number of biophysical mechanisms can account for observed attentional modulation.

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## 102. On the Analysis and Interpretation of Inhomogeneous Quadratic Forms as Receptive Fields

*Pietro Berkes, Laurenz Wiskott*
https://doi.org/10.1162/neco.2006.18.8.1868 | pp. 1868-1895

In this letter, we introduce some mathematical and numerical tools to analyze and interpret inhomogeneous quadratic forms. The resulting characterization is in some aspects similar to that given by experimental studies of cortical cells, making it particularly suitable for application to second-order approximations and theoretical models of physiological receptive fields. We first discuss two ways of analyzing a quadratic form by visualizing the coefficients of its quadratic and linear term directly and by considering the eigenvectors of its quadratic term. We then present an algorithm to compute the optimal excitatory and inhibitory stimuli--those that maximize and minimize the considered quadratic form, respectively, given a fixed energy constraint. The analysis of the optimal stimuli is completed by considering their invariances, which are the transformations to which the quadratic form is most insensitive, and by introducing a test to determine which of these are statistically significant. Next we propose a way to measure the relative contribution of the quadratic and linear term to the total output of the quadratic form. Furthermore, we derive simpler versions of the above techniques in the special case of a quadratic form without linear term. In the final part of the letter, we show that for each quadratic form, it is possible to build an equivalent two-layer neural network, which is compatible with (but more general than) related networks used in some recent articles and with the energy model of complex cells. We show that the neural network is unique only up to an arbitrary orthogonal transformation of the excitatory and inhibitory subunits in the first layer.

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## 103. Comment on “Characterization of Subthreshold Voltage Fluctuations in Neuronal Membranes,” by M. Rudolph and A. Destexhe

*Benjamin Lindner, André Longtin*
https://doi.org/10.1162/neco.2006.18.8.1896 | pp. 1896-1931

In two recent articles, Rudolph and Destexhe (2003, 2005) studied a leaky integrator model (an RC-circuit) driven by correlated ("colored") gaussian conductance noise and Gaussian current noise. In the first article, they derived an expression for the stationary probability density of the membrane voltage; in the second, they modified this expression to cover a larger parameter regime. Here we show by standard analysis of solvable limit cases (white noise limit of additive and multiplicative noise sources; only slow multiplicative noise; only additive noise) and by numerical simulations that their first result does not hold for the general colored-noise case and uncover the errors made in the derivation of a Fokker-Planck equation for the probability density. Furthermore, we demonstrate analytically (including an exact integral expression for the time-dependent mean value of the voltage) and by comparison to simulation results that the extended expression for the probability density works much better but still does not exactly solve the full colored-noise problem. We also show that at stronger synaptic input, the stationary mean value of the linear voltage model may diverge and give an exact condition relating the system parameters for which this takes place.

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## 104. Kernel Projection Classifiers with Suppressing Features of Other Classes

*Yoshikazu Washizawa, Yukihiko Yamashita*
https://doi.org/10.1162/neco.2006.18.8.1932 | pp. 1932-1950

We propose a new classification method based on a kernel technique called suppressed kernel sample space projection classifier (SKSP), which is extended from kernel sample space projection classifier (KSP). In kernel methods, samples are classified after they are mapped from an input space to a high-dimensional space called a feature space. The space that is spanned by samples of a class in a feature space is defined as a kernel sample space. In KSP, an unknown input vector is classified to the class of which projection norm onto the kernel sample space is maximized. KSP can be interpreted as a special type of kernel principal component analysis (KPCA). KPCA is also used in classification problems. However, KSP has more useful properties compared with KPCA, and its accuracy is almost the same as or better than that of KPCA classifier. Since KSP is a single-class classifier, it uses only self-class samples for learning. Thus, for a multiclass classification problem, even though there are very many classes, the computational cost does not change for each class. However, we expect that more useful features can be obtained for classification if samples from other classes are used. By extending KSP to SKSP, the effects of other classes are suppressed, and useful features can be extracted with an oblique projection. Experiments on two-class classification problems, indicate that SKSP shows high accuracy in many classification problems.

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## 105. Implications of Neuronal Diversity on Population Coding

*Maoz Shamir, Haim Sompolinsky*

https://doi.org/10.1162/neco.2006.18.8.1951 | pp. 1951-1986

In many cortical and subcortical areas, neurons are known to modulate their average firing rate in response to certain external stimulus features. It is widely believed that information about the stimulus features is coded by a weighted average of the neural responses. Recent theoretical studies have shown that the information capacity of such a coding scheme is very limited in the presence of the experimentally observed pairwise correlations. However, central to the analysis of these studies was the assumption of a homogeneous population of neurons. Experimental findings show a considerable measure of heterogeneity in the response properties of different neurons. In this study, we investigate the effect of neuronal heterogeneity on the information capacity of a correlated population of neurons. We show that information capacity of a heterogeneous network is not limited by the correlated noise, but scales linearly with the number of cells in the population. This information cannot be extracted by the population vector readout, whose accuracy is greatly suppressed by the correlated noise. On the other hand, we show that an optimal linear readout that takes into account the neuronal heterogeneity can extract most of this information. We study analytically the nature of the dependence of the optimal linear readout weights on the neuronal diversity. We show that simple online learning can generate readout weights with the appropriate dependence on the neuronal diversity, thereby yielding efficient readout.

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## 106. A Neighborhood-Based Enhancement of the Gauss-Newton Bayesian Regularization Training Method

*Miguel Pinzolas, Ana Toledo, Juan Luís Pedreño*

https://doi.org/10.1162/neco.2006.18.8.1987 | pp. 1987-2003

This work develops and tests a neighborhood-based approach to the Gauss-Newton Bayesian regularization training method for feedforward backpropagation networks. The proposed method improves the training efficiency, significantly reducing requirements on memory and computational time while

maintaining the good generalization feature of the original algorithm. This version of the Gauss-Newton Bayesian regularization greatly expands the scope of application of the original method, as it allows training networks up to 100 times larger without losing performance.

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## 107. Exact Simulation of Integrate-and-Fire Models with Synaptic Conductances

*Romain Brette*

https://doi.org/10.1162/neco.2006.18.8.2004 | pp. 2004-2027

Computational neuroscience relies heavily on the simulation of large networks of neuron models. There are essentially two simulation strategies: (1) using an approximation method (e.g., Runge-Kutta) with spike times binned to the time step and (2) calculating spike times exactly in an event-driven fashion. In large networks, the computation time of the best algorithm for either strategy scales linearly with the number of synapses, but each strategy has its own assets and constraints: approximation methods can be applied to any model but are inexact; exact simulation avoids numerical artifacts but is limited to simple models. Previous work has focused on improving the accuracy of approximation methods. In this article, we extend the range of models that can be simulated exactly to a more realistic model: an integrate-and-fire model with exponential synaptic conductances.

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## 108. Bursting Without Slow Kinetics: A Role for a Small World?

*Jie Shao, Tzu-Hsin B. Tsao, Robert J. Butera*

https://doi.org/10.1162/neco.2006.18.9.2029 | pp. 2029-2035

Bursting, a dynamical phenomenon whereby episodes of neural action potentials are punctuated by periodic episodes of inactivity, is ubiquitous in neural systems. Examples include components of the respiratory rhythm generating circuitry in the brain stem, spontaneous activity in the neonatal rat spinal cord, and developing neural networks in the retina of the immature ferret. Bursting can also manifest itself in single neurons. Bursting dynamics require one or more kinetic processes slower than the timescale of the action potentials. Such processes usually manifest themselves in intrinsic ion channel properties, such as slow voltage-dependent gating or calcium-dependent processes, or synaptic mechanisms, such as synaptic depression. In this note, we show rhythmic bursting in a simulated neural network where no such slow processes exist at the cellular or synaptic level. Rather, the existence of rhythmic bursting is critically dependent on the connectivity of the network and manifests itself only when connectivity is characterized as small world. The slow process underlying the timescale of bursting manifests itself as a progressive synchronization of the network within each burst.

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## 109. Error Entropy in Classification Problems: A Univariate Data Analysis

*Luís M. Silva, Carlos S. Felgueiras, Lu■■s A. Alexandre, Joaquim Marques de Sá*

https://doi.org/10.1162/neco.2006.18.9.2036 | pp. 2036-2061

Entropy-based cost functions are enjoying a growing attractiveness in unsupervised and supervised classification tasks. Better performances in terms both of error rate and speed of convergence have been

reported. In this letter, we study the principle of error entropy minimization (EEM) from a theoretical point of view. We use Shannon's entropy and study univariate data splitting in two-class problems. In this setting, the error variable is a discrete random variable, leading to a not too complicated mathematical analysis of the error entropy. We start by showing that for uniformly distributed data, there is equivalence between the EEM split and the optimal classifier. In a more general setting, we prove the necessary conditions for this equivalence and show the existence of class configurations where the optimal classifier corresponds to maximum error entropy. The presented theoretical results provide practical guidelines that are illustrated with a set of experiments with both real and simulated data sets, where the effectiveness of EEM is compared with the usual mean square error minimization.

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## 110. Online Adaptive Decision Trees: Pattern Classification and Function Approximation

*Jayanta Kumar Basak*

https://doi.org/10.1162/neco.2006.18.9.2062 | pp. 2062-2101

Recently we have shown that decision trees can be trained in the online adaptive (OADT) mode (Basak, 2004), leading to better generalization score. OADTs were bottlenecked by the fact that they are able to handle only two-class classification tasks with a given structure. In this article, we provide an architecture based on OADT, ExOADT, which can handle multiclass classification tasks and is able to perform function approximation. ExOADT is structurally similar to OADT extended with a regression layer. We also show that ExOADT is capable not only of adapting the local decision hyperplanes in the nonterminal nodes but also has the potential of smoothly changing the structure of the tree depending on the data samples. We provide the learning rules based on steepest gradient descent for the new model ExOADT. Experimentally we demonstrate the effectiveness of ExOADT in the pattern classification and function approximation tasks. Finally, we briefly discuss the relationship of ExOADT with other classification models.

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## 111. Neuronal Algorithms That Detect the Temporal Order of Events

*Gonzalo G. de Polavieja*

https://doi.org/10.1162/neco.2006.18.9.2102 | pp. 2102-2121

One of the basic operations in sensory processing is the computation of the temporal order of excitation of sensors. Motivated by the discrepancy between models and experiments at high signal contrast, we obtain families of algorithms by solutions of a general set of equations that define temporal order detection as an input-to-output relationship. Delays and nonlinear operations are the basis of all algorithms found, but different algorithmic structures exist when the operations are multiplications, OR gates, different types of AND-NOT logical gates, or concatenated AND-NOT gates. Among others, we obtain the Hassenstein-Reichardt model, a network using a multiplicative operation that has been proposed to explain fly optomotor behavior. We also find extensions of the Barlow-Levick model (based on an AND-NOT gate with delayed inhibition and nondelayed excitation as inputs), originally proposed to explain the bipolar cell response of the rabbit retina to motion stimuli. In the extended models, there are two more steps, another AND-NOT gate, and a subtraction or two subtractions that make the model responsive only to motion. In response to low-contrast inputs, the concatenated AND-NOT gates or the AND-NOT gate followed by a subtraction in these new models act as the multiplicative operation in the Hassenstein-Reichardt model. At

high contrast, the new models behave like the Hassenstein-Reichardt model except that they are independent of contrast as observed experimentally.

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## 112. Excessive Noise Injection Training of Neural Networks for Markerless Tracking in Obscured and Segmented Environments

*Charles P. Unsworth, George Coghill*

https://doi.org/10.1162/neco.2006.18.9.2122 | pp. 2122-2145

In this letter, we demonstrate that the generalization properties of a neural network (NN) can be extended to encompass objects that obscure or segment the original image in its foreground or background. We achieve this by piloting an extension of the noise injection training technique, which we term excessive noise injection (ENI), on a simple feedforward multilayer perceptron (MLP) network with vanilla backward error propagation to achieve this aim. Six tests are reported that show the ability of an NN to distinguish six similar states of motion of a simplified human figure that has become obscured by moving vertical and horizontal bars and random blocks for different levels of obscuration. Four more extensive tests are then reported to determine the bounds of the technique. The results from the ENI network were compared to results from the same NN trained on clean states only. The results pilot strong evidence that it is possible to track a human subject behind objects using this technique, and thus this technique lends itself to a real-time markerless tracking system from a single video stream.

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## 113. Analytical Integrate-and-Fire Neuron Models with Conductance-Based Dynamics for Event-Driven Simulation Strategies

*Michelle Rudolph, Alain Destexhe*

https://doi.org/10.1162/neco.2006.18.9.2146 | pp. 2146-2210

Event-driven simulation strategies were proposed recently to simulate integrate-and-fire (IF) type neuronal models. These strategies can lead to computationally efficient algorithms for simulating large-scale networks of neurons; most important, such approaches are more precise than traditional clock-driven numerical integration approaches because the timing of spikes is treated exactly. The drawback of such event-driven methods is that in order to be efficient, the membrane equations must be solvable analytically, or at least provide simple analytic approximations for the state variables describing the system. This requirement prevents, in general, the use of conductance-based synaptic interactions within the framework of event-driven simulations and, thus, the investigation of network paradigms where synaptic conductances are important. We propose here a number of extensions of the classical leaky IF neuron model involving approximations of the membrane equation with conductance-based synaptic current, which lead to simple analytic expressions for the membrane state, and therefore can be used in the event-driven framework. These conductance-based IF (gIF) models are compared to commonly used models, such as the leaky IF model or biophysical models in which conductances are explicitly integrated. All models are compared with respect to various spiking response properties in the presence of synaptic activity, such as the spontaneous discharge statistics, the temporal precision in resolving synaptic inputs, and gain modulation under in vivo-like synaptic bombardment. Being based on the passive membrane equation with fixed-threshold spike generation, the proposed gIF models are situated in between leaky IF and biophysical models but are much closer to the latter with respect to their dynamic behavior and response characteristics, while still being

nearly as computationally efficient as simple IF neuron models. gIF models should therefore provide a useful tool for efficient and precise simulation of large-scale neuronal networks with realistic, conductance-based synaptic interactions.

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## 114. The Crystallizing Substochastic Sequential Machine Extractor: CrySSMEx

*Henrik Jacobsson*

https://doi.org/10.1162/neco.2006.18.9.2211 | pp. 2211-2255

This letter presents an algorithm, CrySSMEx, for extracting minimal finite state machine descriptions of dynamic systems such as recurrent neural networks. Unlike previous algorithms, CrySSMEx is parameter free and deterministic, and it efficiently generates a series of increasingly refined models. A novel finite stochastic model of dynamic systems and a novel vector quantization function have been developed to take into account the state-space dynamics of the system. The experiments show that (1) extraction from systems that can be described as regular grammars is trivial, (2) extraction from high-dimensional systems is feasible, and (3) extraction of approximative models from chaotic systems is possible. The results are promising, and an analysis of shortcomings suggests some possible further improvements. Some largely overlooked connections, of the field of rule extraction from recurrent neural networks, to other fields are also identified.

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## 115. Assessing Neuronal Coherence with Single-Unit, Multi-Unit, and Local Field Potentials

*Magteld Zeitler, Pascal Fries, Stan Gielen*

https://doi.org/10.1162/neco.2006.18.9.2256 | pp. 2256-2281

The purpose of this study was to obtain a better understanding of neuronal responses to correlated input, in particular focusing on the aspect of synchronization of neuronal activity. The first aim was to obtain an analytical expression for the coherence between the output spike train and correlated input and for the coherence between output spike trains of neurons with correlated input. For Poisson neurons, we could derive that the peak of the coherence between the correlated input and multi-unit activity increases proportionally with the square root of the number of neurons in the multi-unit recording. The coherence between two typical multi-unit recordings (2 to 10 single units) with partially correlated input increases proportionally with the number of units in the multi-unit recordings. The second aim of this study was to investigate to what extent the amplitude and signal-to-noise ratio of the coherence between input and output varied for single-unit versus multi-unit activity and how they are affected by the duration of the recording. The same problem was addressed for the coherence between two single-unit spike series and between two multi-unit spike series. The analytical results for the Poisson neuron and numerical simulations for the conductance-based leaky integrate-and-fire neuron and for the conductance-based Hodgkin-Huxley neuron show that the expectation value of the coherence function does not increase for a longer duration of the recording. The only effect of a longer duration of the spike recording is a reduction of the noise in the coherence function. The results of analytical derivations and computer simulations for model neurons show that the coherence for multi-unit activity is larger than that for single-unit activity. This is in agreement with the results of experimental data obtained from monkey visual cortex (V4). Finally, we show that multitaper

techniques greatly contribute to a more accurate estimate of the coherence by reducing the bias and variance in the coherence estimate.

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## 116. Consistency of Pseudolikelihood Estimation of Fully Visible Boltzmann Machines

*Aapo Hyvärinen*

https://doi.org/10.1162/neco.2006.18.10.2283 | pp. 2283-2292

A Boltzmann machine is a classic model of neural computation, and a number of methods have been proposed for its estimation. Most methods are plagued by either very slow convergence or asymptotic bias in the resulting estimates. Here we consider estimation in the basic case of fully visible Boltzmann machines. We show that the old principle of pseudolikelihood estimation provides an estimator that is computationally very simple yet statistically consistent.

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## 117. Images, Frames, and Connectionist Hierarchies

*Peter Dayan*

https://doi.org/10.1162/neco.2006.18.10.2293 | pp. 2293-2319

The representation of hierarchically structured knowledge in systems using distributed patterns of activity is an abiding concern for the connectionist solution of cognitively rich problems. Here, we use statistical unsupervised learning to consider semantic aspects of structured knowledge representation. We meld unsupervised learning notions formulated for multilinear models with tensor product ideas for representing rich information. We apply the model to images of faces.

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## 118. Properties of Synergies Arising from a Theory of Optimal Motor Behavior

*Manu Chhabra, Robert A. Jacobs*

https://doi.org/10.1162/neco.2006.18.10.2320 | pp. 2320-2342

We consider the properties of motor components, also known as synergies, arising from a computational theory (in the sense of Marr, 1982) of optimal motor behavior. An actor's goals were formalized as cost functions, and the optimal control signals minimizing the cost functions were calculated. Optimal synergies were derived from these optimal control signals using a variant of nonnegative matrix factorization. This was done using two different simulated two--joint arms--an arm controlled directly by torques applied at the joints and an arm in which forces were applied by muscles--and two types of motor tasks-reaching tasks and via-point tasks. Studies of the motor synergies reveal several interesting findings. First, optimal motor actions can be generated by summing a small number of scaled and time-shifted motor synergies, indicating that optimal movements can be planned in a low-dimensional space by using optimal motor synergies as motor primitives or building blocks. Second, some optimal synergies are task independent--they arise regardless of the task context-whereas other synergies are task dependent--they arise in the context of one task but not in the contexts of other tasks. Biological organisms use a combination of task--independent and task--dependent synergies. Our work suggests that this may be an efficient combination for generating

optimal motor actions from motor primitives. Third, optimal motor actions can be rapidly acquired by learning new linear combinations of optimal motor synergies. This result provides further evidence that optimal motor synergies are useful motor primitives. Fourth, synergies with similar properties arise regardless if one uses an arm controlled by torques applied at the joints or an arm controlled by muscles, suggesting that synergies, when considered in "movement space," are more a reflection of task goals and constraints than of fine details of the underlying hardware.

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## 119. Recognition by Variance: Learning Rules for Spatiotemporal Patterns

*Omri Barak, Misha Tsodyks*

https://doi.org/10.1162/neco.2006.18.10.2343 | pp. 2343-2358

Recognizing specific spatiotemporal patterns of activity, which take place at timescales much larger than the synaptic transmission and membrane time constants, is a demand from the nervous system exemplified, for instance, by auditory processing. We consider the total synaptic input that a single readout neuron receives on presentation of spatiotemporal spiking input patterns. Relying on the monotonic relation between the mean and the variance of a neuron's input current and its spiking output, we derive learning rules that increase the variance of the input current evoked by learned patterns relative to that obtained from random background patterns. We demonstrate that the model can successfully recognize a large number of patterns and exhibits a slow deterioration in performance with increasing number of learned patterns. In addition, robustness to time warping of the input patterns is revealed to be an emergent property of the model. Using a leaky integrate-and-fire realization of the readout neuron, we demonstrate that the above results also apply when considering spiking output.

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## 120. Estimating Spiking Irregularities Under Changing Environments

*Keiji Miura, Masato Okada,* ■■■-■■■ ■■■■■

https://doi.org/10.1162/neco.2006.18.10.2359 | pp. 2359-2386

We considered a gamma distribution of interspike intervals as a statistical model for neuronal spike generation. A gamma distribution is a natural extension of the Poisson process taking the effect of a refractory period into account. The model is specified by two parameters: a time-dependent firing rate and a shape parameter that characterizes spiking irregularities of individual neurons. Because the environment changes over time, observed data are generated from a model with a time-dependent firing rate, which is an unknown function. A statistical model with an unknown function is called a semiparametric model and is generally very difficult to solve. We used a novel method of estimating functions in information geometry to estimate the shape parameter without estimating the unknown function. We obtained an optimal estimating function analytically for the shape parameter independent of the functional form of the firing rate. This estimation is efficient without Fisher information loss and better than maximum likelihood estimation. We suggest a measure of spiking irregularity based on the estimating function, which may be useful for characterizing individual neurons in changing environments.

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## 121. Spatiotemporal Structure in Large Neuronal Networks Detected from Cross-Correlation

*Gaby Schneider, Martha N. Havenith, Danko Nikoli■*

https://doi.org/10.1162/neco.2006.18.10.2387 | pp. 2387-2413

The analysis of neuronal information involves the detection of spatiotemporal relations between neuronal discharges. We propose a method that is based on the positions (phase offsets) of the central peaks obtained from pairwise cross-correlation histograms. Data complexity is reduced to a one-dimensional representation by using redundancies in the measured phase offsets such that each unit is assigned a "preferred firing time" relative to the other units in the group. We propose two procedures to examine the applicability of this method to experimental data sets. In addition, we propose methods that help the investigation of dynamical changes in the preferred firing times of the units. All methods are applied to a sample data set obtained from cat visual cortex.

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## 122. Stable Competitive Dynamics Emerge from Multispike Interactions in a Stochastic Model of Spike-Timing-Dependent Plasticity

*Peter A. Appleby, Terry Elliott*

https://doi.org/10.1162/neco.2006.18.10.2414 | pp. 2414-2464

In earlier work we presented a stochastic model of spike-timing-dependent plasticity (STDP) in which STDP emerges only at the level of temporal or spatial synaptic ensembles. We derived the two-spike interaction function from this model and showed that it exhibits an STDP-like form. Here, we extend this work by examining the general n-spike interaction functions that may be derived from the model. A comparison between the two-spike interaction function and the higher-order interaction functions reveals profound differences. In particular, we show that the two-spike interaction function cannot support stable, competitive synaptic plasticity, such as that seen during neuronal development, without including modifications designed specifically to stabilize its behavior. In contrast, we show that all the higher-order interaction functions exhibit a fixed-point structure consistent with the presence of competitive synaptic dynamics. This difference originates in the unification of our proposed "switch" mechanism for synaptic plasticity, coupling synaptic depression and synaptic potentiation processes together. While three or more spikes are required to probe this coupling, two spikes can never do so. We conclude that this coupling is critical to the presence of competitive dynamics and that multispike interactions are therefore vital to understanding synaptic competition.

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## 123. A State-Space Analysis for Reconstruction of Goal-Directed Movements Using Neural Signals

*Lakshminarayan Srinivasan, Uri T. Eden, Alan S. Willsky, Emery N. Brown*

https://doi.org/10.1162/neco.2006.18.10.2465 | pp. 2465-2494

The execution of reaching movements involves the coordinated activity of multiple brain regions that relate variously to the desired target and a path of arm states to achieve that target. These arm states may represent positions, velocities, torques, or other quantities. Estimation has been previously applied to neural activity in reconstructing the target separately from the path. However, the target and path are not

independent. Because arm movements are limited by finite muscle contractility, knowledge of the target constrains the path of states that leads to the target. In this letter, we derive and illustrate a state equation to capture this basic dependency between target and path. The solution is described for discrete-time linear systems and gaussian increments with known target arrival time. The resulting analysis enables the use of estimation to study how brain regions that relate variously to target and path together specify a trajectory. The corresponding reconstruction procedure may also be useful in brain-driven prosthetic devices to generate control signals for goal-directed movements.

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## 124. What Is the Relation Between Slow Feature Analysis and Independent Component Analysis?

*Tobias Blaschke, Pietro Berkes, Laurenz Wiskott*

https://doi.org/10.1162/neco.2006.18.10.2495 | pp. 2495-2508

We present an analytical comparison between linear slow feature analysis and second-order independent component analysis, and show that in the case of one time delay, the two approaches are equivalent. We also consider the case of several time delays and discuss two possible extensions of slow feature analysis.

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## 125. Nonlocal Estimation of Manifold Structure

*Yoshua Bengio, Martin Monperrus, Hugo Larochelle*

https://doi.org/10.1162/neco.2006.18.10.2509 | pp. 2509-2528

We claim and present arguments to the effect that a large class of manifold learning algorithms that are essentially local and can be framed as kernel learning algorithms will suffer from the curse of dimensionality, at the dimension of the true underlying manifold. This observation invites an exploration of nonlocal manifold learning algorithms that attempt to discover shared structure in the tangent planes at different positions. A training criterion for such an algorithm is proposed, and experiments estimating a tangent plane prediction function are presented, showing its advantages with respect to local manifold learning algorithms: it is able to generalize very far from training data (on learning handwritten character image rotations), where local nonparametric methods fail.

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## 126. Dynamics and Topographic Organization of Recursive Self-Organizing Maps

*Peter Ti■o, Igor Farkaš, Jort van Mourik*

https://doi.org/10.1162/neco.2006.18.10.2529 | pp. 2529-2567

Recently there has been an outburst of interest in extending topographic maps of vectorial data to more general data structures, such as sequences or trees. However, there is no general consensus as to how best to process sequences using topographic maps, and this topic remains an active focus of neurocomputational research. The representational capabilities and internal representations of the models are not well understood. Here, we rigorously analyze a generalization of the self-organizing map (SOM) for processing sequential data, recursive SOM(RecSOM) (Voegtlin, 2002), as a nonautonomous dynamical system consisting of a set of fixed input maps. We argue that contractive fixed-input maps are likely to

produce Markovian organizations of receptive fields on the RecSOM map. We derive bounds on parameter beta (weighting the importance of importing past information when processing sequences) under which contractiveness of the fixed-input maps is guaranteed. Some generalizations of SOM contain a dynamic module responsible for processing temporal contexts as an integral part of the model. We show that Markovian topographic maps of sequential data can be produced using a simple fixed (nonadaptable) dynamic module externally feeding a standard topographic model designed to process static vectorial data of fixed dimensionality (e.g., SOM). However, by allowing trainable feedback connections, one can obtain Markovian maps with superior memory depth and topography preservation. We elaborate on the importance of non-Markovian organizations in topographic maps of sequential data.

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## 127. On the Consistency of the Blocked Neural Network Estimator in Time Series Analysis

*Alex Sarishvili, C-H Andersson, Jürgen Franke, Gerald Kroisandt*

https://doi.org/10.1162/neco.2006.18.10.2568 | pp. 2568-2581

We describe a nonlinear regression problem, where the regression functions have an additive structure and the dependent variable is a one-dimensional time series. Multivariate time series with unknown time delay operators are used as independent variables. By fitting a feedforward neural network with block structure to the data, we estimated the additive regression function and, parallel to this, the time lags. We present the consistency proof of neural network weights estimator and the time lag estimator independently from each other. In the practical part of the article, we present the useful feature of blocked neural networks to estimate the relevance measures of each input variable in a simple way. Furthermore, we propose an approach to solve the well-known variable selection problem for the class of nonlinear multivariate beta-mixing time series models considered here. Finally, we apply the methodology to an artificial example.

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## 128. Erratum

*(authors unknown)*

https://doi.org/10.1162/neco.2006.18.10.2582 | pp. 2582-2582

No abstract available.

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## 129. Spike Count Correlation Increases with Length of Time Interval in the Presence of Trial-to-Trial Variation

*Robert E. Kass, Valérie Ventura*

https://doi.org/10.1162/neco.2006.18.11.2583 | pp. 2583-2591

It has been observed that spike count correlation between two simultaneously recorded neurons often increases with the length of time interval examined. Under simple assumptions that are roughly consistent with much experimental data, we show that this phenomenon may be explained as being due to excess trial-to-trial variation. The resulting formula for the correlation is able to predict the observed correlation of two neurons recorded from primary visual cortex as a function of interval length.

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## 130. The Spike-Triggered Average of the Integrate-and-Fire Cell Driven by Gaussian White Noise

*Liam Paninski*

https://doi.org/10.1162/neco.2006.18.11.2592 | pp. 2592-2616

We compute the exact spike-triggered average (STA) of the voltage for the nonleaky integrate-and-fire (IF) cell in continuous time, driven by gaussian white noise. The computation is based on techniques from the theory of renewal processes and continuous-time hidden Markov processes (e.g., the backward and forward Fokker-Planck partial differential equations associated with first-passage time densities). From the STA voltage, it is straightforward to derive the STA input current. The theory also gives an explicit asymptotic approximation for the STA of the leaky IF cell, valid in the low-noise regime sigma --> 0. We consider both the STA and the conditional average voltage given an observed spike "doublet" event, that is, two spikes separated by some fixed period of silence. In each case, we find that the STA as a function of time-preceding-spike, tau, has a square root singularity as tau approaches zero from below and scales linearly with the scale of injected noise current. We close by briefly examining the discrete-time case, where similar phenomena are observed.

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## 131. Low-Dimensional Maps Encoding Dynamics in Entorhinal Cortex and Hippocampus

*Dmitri D. Pervouchine, Théoden I. Netoff, Horacio G. Rotstein, John A. White, Mark O. Cunningham, Miles A. Whittington, Nancy Kopell*

https://doi.org/10.1162/neco.2006.18.11.2617 | pp. 2617-2650

Cells that produce intrinsic theta oscillations often contain the hyperpolarization-activated current I(h). In this article, we use models and dynamic clamp experiments to investigate the synchronization properties of two such cells (stellate cells of the entorhinal cortex and O-LM cells of the hippocampus) in networks with fast-spiking (FS) interneurons. The model we use for stellate cells and O-LM cells is the same, but the stellate cells are excitatory and the O-LM cells are inhibitory, with inhibitory postsynaptic potential considerably longer than those from FS interneurons. We use spike time response curve methods (STRC), expanding that technique to three-cell networks and giving two different ways in which the analysis of the three-cell network reduces to that of a two-cell network. We show that adding FS cells to a network of stellate cells can desynchronize the stellate cells, while adding them to a network of O-LM cells can synchronize the O-LM cells. These synchronization and desynchronization properties critically depend on I(h). The analysis of the deterministic system allows us to understand some effects of noise on the phase relationships in the stellate networks. The dynamic clamp experiments use biophysical stellate cells and in silico FS cells, with connections that mimic excitation or inhibition, the latter with decay times associated with FS cells or O-LM cells. The results obtained in the dynamic clamp experiments are in a good agreement with the analytical framework.

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## 132. Programmable Logic Construction Kits for Hyper-Real-Time Neuronal Modeling

*Rubén Guerrero■Rivera, Abigail Morrison, Markus Diesmann, Tim Pearce*

https://doi.org/10.1162/neco.2006.18.11.2651 | pp. 2651-2679

Programmable logic designs are presented that achieve exact integration of leaky integrate-and-fire soma and dynamical synapse neuronal models and incorporate spike-time dependent plasticity and axonal delays. Highly accurate numerical performance has been achieved by modifying simpler forward-Euler-based circuitry requiring minimal circuit allocation, which, as we show, behaves equivalently to exact integration. These designs have been implemented and simulated at the behavioral and physical device levels, demonstrating close agreement with both numerical and analytical results. By exploiting finely grained parallelism and single clock cycle numerical iteration, these designs achieve simulation speeds at least five orders of magnitude faster than the nervous system, termed here hyper-real-time operation, when deployed on commercially available field-programmable gate array (FPGA) devices. Taken together, our designs form a programmable logic construction kit of commonly used neuronal model elements that supports the building of large and complex architectures of spiking neuron networks for real-time neuromorphic implementation, neurophysiological interfacing, or efficient parameter space investigations.

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## 133. Soft Mixer Assignment in a Hierarchical Generative Model of Natural Scene Statistics

*Odelia Schwartz, Terrence J. Sejnowski, Peter Dayan*

https://doi.org/10.1162/neco.2006.18.11.2680 | pp. 2680-2718

Gaussian scale mixture models offer a top-down description of signal generation that captures key bottom-up statistical characteristics of filter responses to images. However, the pattern of dependence among the filters for this class of models is prespecified. We propose a novel extension to the gaussian scale mixture model that learns the pattern of dependence from observed inputs and thereby induces a hierarchical representation of these inputs. Specifically, we propose that inputs are generated by gaussian variables (modeling local filter structure), multiplied by a mixer variable that is assigned probabilistically to each input from a set of possible mixers. We demonstrate inference of both components of the generative model, for synthesized data and for different classes of natural images, such as a generic ensemble and faces. For natural images, the mixer variable assignments show invariances resembling those of complex cells in visual cortex; the statistics of the gaussian components of the model are in accord with the outputs of divisive normalization models. We also show how our model helps interrelate a wide range of models of image statistics and cortical processing.

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## 134. The Scaling of Winner-Takes-All Accuracy with Population Size

*Maoz Shamir*

https://doi.org/10.1162/neco.2006.18.11.2719 | pp. 2719-2729

Empirical studies seem to support conflicting hypotheses with regard to the nature of the neural code. While some studies highlight the role of a distributed population code, others emphasize the possibility of a "single-best-cell" readout. One particularly interesting example of single-best-cell readout is provided by the winner-takes-all (WTA) approach. According to the WTA, every cell is characterized by one particular preferred stimulus, to which it responds maximally. The WTA estimate for the stimulus is defined as the preferred stimulus of the cell with the strongest response. From a theoretical point of view, not much is known about the efficiency of single-best-cell readout mechanisms, in contrast to the considerable existing theoretical knowledge on the efficiency of distributed population codes. In this work, we provide a basic

theoretical framework for investigating single-best-cell readout mechanisms. We study the accuracy of the WTA readout. In particular, we are interested in how the WTA accuracy scales with the number of cells in the population. Using this framework, we show that for large neuronal populations, the WTA accuracy is dominated by the tail of the single-cell-response distribution. Furthermore, we find that although the WTA accuracy does improve when larger populations are considered, this improvement is extremely weak compared to other types of population codes. More precisely, we show that while the accuracy of a linear readout scales linearly with the population size, the accuracy of the WTA readout scales logarithmically with the number of cells in the population.

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## 135. An Extended EM Algorithm for Joint Feature Extraction and Classification in Brain-Computer Interfaces

*Yuanqing Li, Cuntai Guan*

https://doi.org/10.1162/neco.2006.18.11.2730 | pp. 2730-2761

For many electroencephalogram (EEG)-based brain-computer interfaces (BCIs), a tedious and time-consuming training process is needed to set parameters. In BCI Competition 2005, reducing the training process was explicitly proposed as a task. Furthermore, an effective BCI system needs to be adaptive to dynamic variations of brain signals; that is, its parameters need to be adjusted online. In this article, we introduce an extended expectation maximization (EM) algorithm, where the extraction and classification of common spatial pattern (CSP) features are performed jointly and iteratively. In each iteration, the training data set is updated using all or part of the test data and the labels predicted in the previous iteration. Based on the updated training data set, the CSP features are reextracted and classified using a standard EM algorithm. Since the training data set is updated frequently, the initial training data set can be small (semi-supervised case) or null (unsupervised case). During the above iterations, the parameters of the Bayes classifier and the CSP transformation matrix are also updated concurrently. In online situations, we can still run the training process to adjust the system parameters using unlabeled data while a subject is using the BCI system. The effectiveness of the algorithm depends on the robustness of CSP feature to noise and iteration convergence, which are discussed in this article. Our proposed approach has been applied to data set IVa of BCI Competition 2005. The data analysis results show that we can obtain satisfying prediction accuracy using our algorithm in the semisupervised and unsupervised cases. The convergence of the algorithm and robustness of CSP feature are also demonstrated in our data analysis.

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## 136. On the Consistency of Bayesian Variable Selection for High Dimensional Binary Regression and Classification

*Wenxin Jiang*

https://doi.org/10.1162/neco.2006.18.11.2762 | pp. 2762-2776

Modern data mining and bioinformatics have presented an important playground for statistical learning techniques, where the number of input variables is possibly much larger than the sample size of the training data. In supervised learning, logistic regression or probit regression can be used to model a binary output and form perceptron classification rules based on Bayesian inference. We use a prior to select a limited number of candidate variables to enter the model, applying a popular method with selection indicators. We

show that this approach can induce posterior estimates of the regression functions that are consistently estimating the truth, if the true regression model is sparse in the sense that the aggregated size of the regression coefficients are bounded. The estimated regression functions therefore can also produce consistent classifiers that are asymptotically optimal for predicting future binary outputs. These provide theoretical justifications for some recent empirical successes in microarray data analysis.

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## 137. A Maximum Likelihood Approach to Density Estimation with Semidefinite Programming

*Tadayoshi Fushiki, Shingo Horiuchi, Takashi Tsuchiya*

https://doi.org/10.1162/neco.2006.18.11.2777 | pp. 2777-2812

Density estimation plays an important and fundamental role in pattern recognition, machine learning, and statistics. In this article, we develop a parametric approach to univariate (or low-dimensional) density estimation based on semidefinite programming (SDP). Our density model is expressed as the product of a nonnegative polynomial and a base density such as normal distribution, exponential distribution, and uniform distribution. When the base density is specified, the maximum likelihood estimation of the polynomial is formulated as a variant of SDP that is solved in polynomial time with the interior point methods. Since the base density typically contains just one or two parameters, computation of the maximum likelihood estimate reduces to a one- or two-dimensional easy optimization problem with this use of SDP. Thus, the rigorous maximum likelihood estimate can be computed in our approach. Furthermore, such conditions as symmetry and unimodality of the density function can be easily handled within this framework. AIC is used to choose the best model. Through applications to several instances, we demonstrate flexibility of the model and performance of the proposed procedure. Combination with a mixture approach is also presented. The proposed approach has possible other applications beyond density estimation. This point is clarified through an application to the maximum likelihood estimation of the intensity function of a nonstationary Poisson process.

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## 138. Piecewise-Linear Neural Networks and Their Relationship to Rule Extraction from Data

*Martin Hole■a*

https://doi.org/10.1162/neco.2006.18.11.2813 | pp. 2813-2853

This article addresses the topic of extracting logical rules from data by means of artificial neural networks. The approach based on piecewise linear neural networks is revisited, which has already been used for the extraction of Boolean rules in the past, and it is shown that this approach can be important also for the extraction of fuzzy rules. Two important theoretical properties of piecewise-linear neural networks are proved, allowing an elaboration of the basic ideas of the approach into several variants of an algorithm for the extraction of Boolean rules. That algorithm has already been used in two real-world applications. Finally, a connection to the extraction of rules of the ■ukasiewicz logic is established, relying on recent results about rational McNaughton functions. Based on one of the constructive proofs of the McNaughton theorem, an algorithm is formulated that in principle allows extracting a particular kind of formulas of the ■ukasiewicz predicate logic from piecewise-linear neural networks trained with rational data.

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## 139. Computation of Madalines' Sensitivity to Input and Weight Perturbations

*Yingfeng Wang, Xiaoqin Zeng, Daniel Yeung, Zhihang Peng*

https://doi.org/10.1162/neco.2006.18.11.2854 | pp. 2854-2877

The sensitivity of a neural network's output to its input and weight perturbations is an important measure for evaluating the network's performance. In this letter, we propose an approach to quantify the sensitivity of Madalines. The sensitivity is defined as the probability of output deviation due to input and weight perturbations with respect to overall input patterns. Based on the structural characteristics of Madalines, a bottom-up strategy is followed, along which the sensitivity of single neurons, that is, Adalines, is considered first and then the sensitivity of the entire Madaline network. By means of probability theory, an analytical formula is derived for the calculation of Adalines' sensitivity, and an algorithm is designed for the computation of Madalines' sensitivity. Computer simulations are run to verify the effectiveness of the formula and algorithm. The simulation results are in good agreement with the theoretical results.

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## 140. Short-Term Synaptic Plasticity Can Enhance Weak Signal Detectability in Nonrenewal Spike Trains

*Niklas Lüdtke, Mark Nelson*

https://doi.org/10.1162/neco.2006.18.12.2879 | pp. 2879-2916

We study the encoding of weak signals in spike trains with interspike interval (ISI) correlations and the signals' subsequent detection in sensory neurons. Motivated by the observation of negative ISI correlations in auditory and electrosensory afferents, we assess the theoretical performance limits of an individual detector neuron receiving a weak signal distributed across multiple afferent inputs. We assess the functional role of ISI correlations in the detection process using statistical detection theory and derive two sequential likelihood ratio detector models: one for afferents with renewal statistics; the other for afferents with negatively correlated ISIs. We suggest a mechanism that might enable sensory neurons to implicitly compute conditional probabilities of presynaptic spikes by means of short-term synaptic plasticity. We demonstrate how this mechanism can enhance a postsynaptic neuron's sensitivity to weak signals by exploiting the correlation structure of the input spike trains. Our model not only captures fundamental aspects of early electrosensory signal processing in weakly electric fish, but may also bear relevance to the mammalian auditory system and other sensory modalities.

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## 141. On the Use of Analytical Expressions for the Voltage Distribution to Analyze Intracellular Recordings

*Michelle Rudolph, Alain Destexhe*

https://doi.org/10.1162/neco.2006.18.12.2917 | pp. 2917-2922

Different analytical expressions for the membrane potential distribution of membranes subject to synaptic noise have been proposed and can be very helpful in analyzing experimental data. However, all of these expressions are either approximations or limit cases, and it is not clear how they compare and which expression should be used in a given situation. In this note, we provide a comparison of the different approximations available, with an aim of delineating which expression is most suitable for analyzing experimental data.

## 142. A Distributed Computing Tool for Generating Neural Simulation Databases

*Robert Calin■Jageman, Paul S. Katz*

https://doi.org/10.1162/neco.2006.18.12.2923 | pp. 2923-2927

After developing a model neuron or network, it is important to systematically explore its behavior across a wide range of parameter values or experimental conditions, or both. However, compiling a very large set of simulation runs is challenging because it typically requires both access to and expertise with high-performance computing facilities. To lower the barrier for large-scale model analysis, we have developed NeuronPM, a client/server application that creates a "screen-saver" cluster for running simulations in NEURON (Hines & Carnevale, 1997). NeuronPM provides a user-friendly way to use existing computing resources to catalog the performance of a neural simulation across a wide range of parameter values and experimental conditions. The NeuronPM client is a Windows-based screen saver, and the NeuronPM server can be hosted on any Apache/PHP/MySQL server. During idle time, the client retrieves model files and work assignments from the server, invokes NEURON to run the simulation, and returns results to the server. Administrative panels make it simple to upload model files, define the parameters and conditions to vary, and then monitor client status and work progress. NeuronPM is open-source freeware and is available for download at http://neuronpm.homeip.net . It is a useful entry-level tool for systematically analyzing complex neuron and network simulations.

## 143. Kernel Least-Squares Models Using Updates of the Pseudoinverse

*Edin Andelic, Martin Schafföner, Marcel Katz, Sven E. Krüger, Andreas Wendemuth*

https://doi.org/10.1162/neco.2006.18.12.2928 | pp. 2928-2935

Sparse nonlinear classification and regression models in reproducing kernel Hilbert spaces (RKHSs) are considered. The use of Mercer kernels and the square loss function gives rise to an overdetermined linear least-squares problem in the corresponding RKHS. When we apply a greedy forward selection scheme, the least-squares problem may be solved by an order-recursive update of the pseudoinverse in each iteration step. The computational time is linear with respect to the number of the selected training samples.

## 144. Learning Tetris Using the Noisy Cross-Entropy Method

*István Szita, András Lörincz*

https://doi.org/10.1162/neco.2006.18.12.2936 | pp. 2936-2941

The cross-entropy method is an efficient and general optimization algorithm. However, its applicability in reinforcement learning (RL) seems to be limited because it often converges to suboptimal policies. We apply noise for preventing early convergence of the cross-entropy method, using Tetris, a computer game, for demonstration. The resulting policy outperforms previous RL algorithms by almost two orders of magnitude.

## 145. A Spiking Neuron Model of Cortical Correlates of Sensorineural Hearing Loss: Spontaneous Firing, Synchrony, and Tinnitus

*Melissa Dominguez, Suzanna Becker, Ian C. Bruce, Heather L. Read*

https://doi.org/10.1162/neco.2006.18.12.2942 | pp. 2942-2958

Hearing loss due to peripheral damage is associated with cochlear hair cell damage or loss and some retrograde degeneration of auditory nerve fibers. Surviving auditory nerve fibers in the impaired region exhibit elevated and broadened frequency tuning, and the cochleotopic representation of broadband stimuli such as speech is distorted. In impaired cortical regions, increased tuning to frequencies near the edge of the hearing loss coupled with increased spontaneous and synchronous firing is observed. Tinnitus, an auditory percept in the absence of sensory input, may arise under these circumstances as a result of plastic reorganization in the auditory cortex. We present a spiking neuron model of auditory cortex that captures several key features of cortical organization. A key assumption in the model is that in response to reduced afferent excitatory input in the damaged region, a compensatory change in the connection strengths of lateral excitatory and inhibitory connections occurs. These changes allow the model to capture some of the cortical correlates of sensorineural hearing loss, including changes in spontaneous firing and synchrony; these phenomena may explain central tinnitus. This model may also be useful for evaluating procedures designed to segregate synchronous activity underlying tinnitus and for evaluating adaptive hearing devices that compensate for selective hearing loss.

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## 146. Event-Driven Simulation Scheme for Spiking Neural Networks Using Lookup Tables to Characterize Neuronal Dynamics

*Eduardo Ros, Richard R. Carrillo, Eva M. Ortigosa, Boris Barbour, Rodrigo Agís*

https://doi.org/10.1162/neco.2006.18.12.2959 | pp. 2959-2993

Nearly all neuronal information processing and interneuronal communication in the brain involves action potentials, or spikes, which drive the short-term synaptic dynamics of neurons, but also their long-term dynamics, via synaptic plasticity. In many brain structures, action potential activity is considered to be sparse. This sparseness of activity has been exploited to reduce the computational cost of large-scale network simulations, through the development of event-driven simulation schemes. However, existing event-driven simulations schemes use extremely simplified neuronal models. Here, we implement and evaluate critically an event-driven algorithm (ED-LUT) that uses precalculated look-up tables to characterize synaptic and neuronal dynamics. This approach enables the use of more complex (and realistic) neuronal models or data in representing the neurons, while retaining the advantage of high-speed simulation. We demonstrate the method's application for neurons containing exponential synaptic conductances, thereby implementing shunting inhibition, a phenomenon that is critical to cellular computation. We also introduce an improved two-stage event-queue algorithm, which allows the simulations to scale efficiently to highly connected networks with arbitrary propagation delays. Finally, the scheme readily accommodates implementation of synaptic plasticity mechanisms that depend on spike timing, enabling future simulations to explore issues of long-term learning and adaptation in large-scale networks.

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## 147. On the Computational Power of Threshold Circuits with Sparse Activity

*Kei Uchizawa, Rodney J. Douglas, Wolfgang Maass*
https://doi.org/10.1162/neco.2006.18.12.2994 | pp. 2994-3008

Circuits composed of threshold gates (McCulloch-Pitts neurons, or perceptrons) are simplified models of neural circuits with the advantage that they are theoretically more tractable than their biological counterparts. However, when such threshold circuits are designed to perform a specific computational task, they usually differ in one important respect from computations in the brain: they require very high activity. On average every second threshold gate fires (sets a 1 as output) during a computation. By contrast, the activity of neurons in the brain is much sparser, with only about 1% of neurons firing. This mismatch between threshold and neuronal circuits is due to the particular complexity measures (circuit size and circuit depth) that have been minimized in previous threshold circuit constructions. In this letter, we investigate a new complexity measure for threshold circuits, energy complexity, whose minimization yields computations with sparse activity. We prove that all computations by threshold circuits of polynomial size with entropy O(log n) can be restructured so that their energy complexity is reduced to a level near the entropy of circuit states. This entropy of circuit states is a novel circuit complexity measure, which is of interest not only in the context of threshold circuits but for circuit complexity in general. As an example of how this measure can be applied, we show that any polynomial size threshold circuit with entropy O(log n) can be simulated by a polynomial size threshold circuit of depth 3. Our results demonstrate that the structure of circuits that result from a minimization of their energy complexity is quite different from the structure that results from a minimization of previously considered complexity measures, and potentially closer to the structure of neural circuits in the nervous system. In particular, different pathways are activated in these circuits for different classes of inputs. This letter shows that such circuits with sparse activity have a surprisingly large computational power.

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## 148. Parameter Space Structure of Continuous-Time Recurrent Neural Networks

*Randall D. Beer*
https://doi.org/10.1162/neco.2006.18.12.3009 | pp. 3009-3051

A fundamental challenge for any general theory of neural circuits is how to characterize the structure of the space of all possible circuits over a given model neuron. As a first step in this direction, this letter begins a systematic study of the global parameter space structure of continuous-time recurrent neural networks (CTRNNs), a class of neural models that is simple but dynamically universal. First, we explicitly compute the local bifurcation manifolds of CTRNNs. We then visualize the structure of these manifolds in net input space for small circuits. These visualizations reveal a set of extremal saddle node bifurcation manifolds that divide CTRNN parameter space into regions of dynamics with different effective dimensionality. Next, we completely characterize the combinatorics and geometry of an asymptotically exact approximation to these regions for circuits of arbitrary size. Finally, we show how these regions can be used to calculate estimates of the probability of encountering different kinds of dynamics in CTRNN parameter space.

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## 149. Bifurcation Analysis of Jansen's Neural Mass Model

*François Grimbert, Olivier Faugeras*
https://doi.org/10.1162/neco.2006.18.12.3052 | pp. 3052-3068

We present a mathematical model of a neural mass developed by a number of people, including Lopes da Silva and Jansen. This model features three interacting populations of cortical neurons and is described by a six-dimensional nonlinear dynamical system. We address some aspects of its behavior through a bifurcation analysis with respect to the input parameter of the system. This leads to a compact description of the oscillatory behaviors observed in Jansen and Rit (1995) (alpha activity) and Wendling, Bellanger, Bartolomei, and Chauvel (2000) (spike-like epileptic activity). In the case of small or slow variation of the input, the model can even be described as a binary unit. Again using the bifurcation framework, we discuss the influence of other parameters of the system on the behavior of the neural mass model.

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## 150. A Model for Perceptual Averaging and Stochastic Bistable Behavior and the Role of Voluntary Control

*Ansgar Koene*

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We combine population coding, winner-take-all competition, and differentiated inhibitory feedback to model the process by which information from different, continuously variable signals is integrated for perceptual awareness. We focus on "slant rivalry," where binocular disparity is in conflict with monocular perspective in specifying surface slant. Using a robust single parameter set, our model successfully replicates three key experimental results: (1) transition from signal averaging to bistability with increasing signal conflict, (2) change in perceptual reversal rates as a function of signal conflict, and (3) a shift in the distribution of percept durations through voluntary control exertion. Voluntary control is implemented through the use of a single top-down bias input. The transition from signal averaging to bistability arises as a natural consequence of combining population coding and wide receptive fields, common to higher cortical areas. The model architecture does not contain any assumption that would limit it to this particular example of stimulus rivalry. An emergent physiological interpretation is that differentiated inhibitory feedback may play an important role for increasing percept stability without reducing sensitivity to large stimulus changes, which for bistable conditions leads to increased alternation rate as a function of signal conflict.

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## 151. A Unifying View of Wiener and Volterra Theory and Polynomial Kernel Regression

*Matthias Franz, Bernhard Schölkopf*

https://doi.org/10.1162/neco.2006.18.12.3097 | pp. 3097-3118

Volterra and Wiener series are perhaps the best-understood nonlinear system representations in signal processing. Although both approaches have enjoyed a certain popularity in the past, their application has been limited to rather low-dimensional and weakly nonlinear systems due to the exponential growth of the number of terms that have to be estimated. We show that Volterra and Wiener series can be represented implicitly as elements of a reproducing kernel Hilbert space by using polynomial kernels. The estimation complexity of the implicit representation is linear in the input dimensionality and independent of the degree of nonlinearity. Experiments show performance advantages in terms of convergence, interpretability, and system sizes that can be handled.

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## 152. An Upper Bound on the Minimum Number of Monomials Required to Separate Dichotomies of {–1, 1}<sup>n</sup>

*Erhan Öztop*

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It is known that any dichotomy of {-1, 1}n can be learned (separated) with a higher-order neuron (polynomial function) with 2n inputs (monomials). In general, less than 2n monomials are sufficient to solve a given dichotomy. In spite of the efforts to develop algorithms for finding solutions with fewer monomials, there have been relatively fewer studies investigating maximum density (Pi(n)), the minimum number of monomials that would suffice to separate an arbitrary dichotomy of {-1, 1}n . This article derives a theoretical (upper) bound for this quantity, superseding previously known bounds. The main theorem here states that for any binary classification problem in {-1, 1}n (n > 1), one can always find a polynomial function solution with 2n -2n/4 or fewer monomials. In particular, any dichotomy of {-1, 1}n can be learned by a higher-order neuron with a fan-in of 2n -2n/4 or less. With this result, for the first time, a deterministic ratio bound independent of n is established as Pi(n)/2n < or = 0 75. The main theorem is constructive, so it provides a deterministic algorithm for achieving the theoretical result. The study presented provides the basic mathematical tools and forms the basis for further analyses that may have implications for neural computation mechanisms employed in the cerebral cortex.

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## 153. Index

*(authors unknown)*

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No abstract available.